The National Higher School of
Artificial Intelligence
المدرسة الوطنية العليا للذكاء الاصطناعي

Course
« *Computer Vision*»

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2025-2026

Image Filtering in the Frequency Domain

- During the past century, and especially in the past 50 years, entire industries and academic disciplines have flourished as a result of Fourier's ideas.
- The "discovery" of a fast Fourier transform (FFT) algorithm in the early 1960s revolutionized the field of signal processing.
- Today, we can say that there is no discipline of science or engineering that has not been profoundly impacted by the Fourier transform.
- The goal of this chapter is to give a working knowledge of how the Fourier transform and the frequency domain can be used for image filtering.

Image Filtering in the Frequency Domain

Before any math:

- The frequency domain describes **how fast image intensities change in space.**
- NOT:
  - “sinusoids”
  - “Fourier formulas”
- **Intuitive definition of “frequency” in an image:**

Frequency describes how rapidly pixel values change across the image.

Image Filtering in the Frequency Domain

Why do we need frequencies in image analysis/processing?

Filtering becomes easier

- In spatial domain:
  - Convolution is expensive
- In frequency domain:
  - Filtering = simple multiplication

We can separate noise and signal:

- Noise → high frequencies
- Structures → low frequencies

- **Convolution theorem**

Convolution in spatial domain =

Multiplication in frequency domain.

```
Spatial domain:
Image ⊗ Filter  (convolution)

Frequency domain:
Spectrum × FilterSpectrum  (simple multiplication)
```

Image Filtering in the Frequency Domain

Very simple intuitive examples:

- **LOW frequency = smooth regions**
  - Sky
  - Large uniform areas
  - Slowly varying intensity

- **HIGH frequency = rapid changes**
  - Edges
  - Fine textures
  - Noise

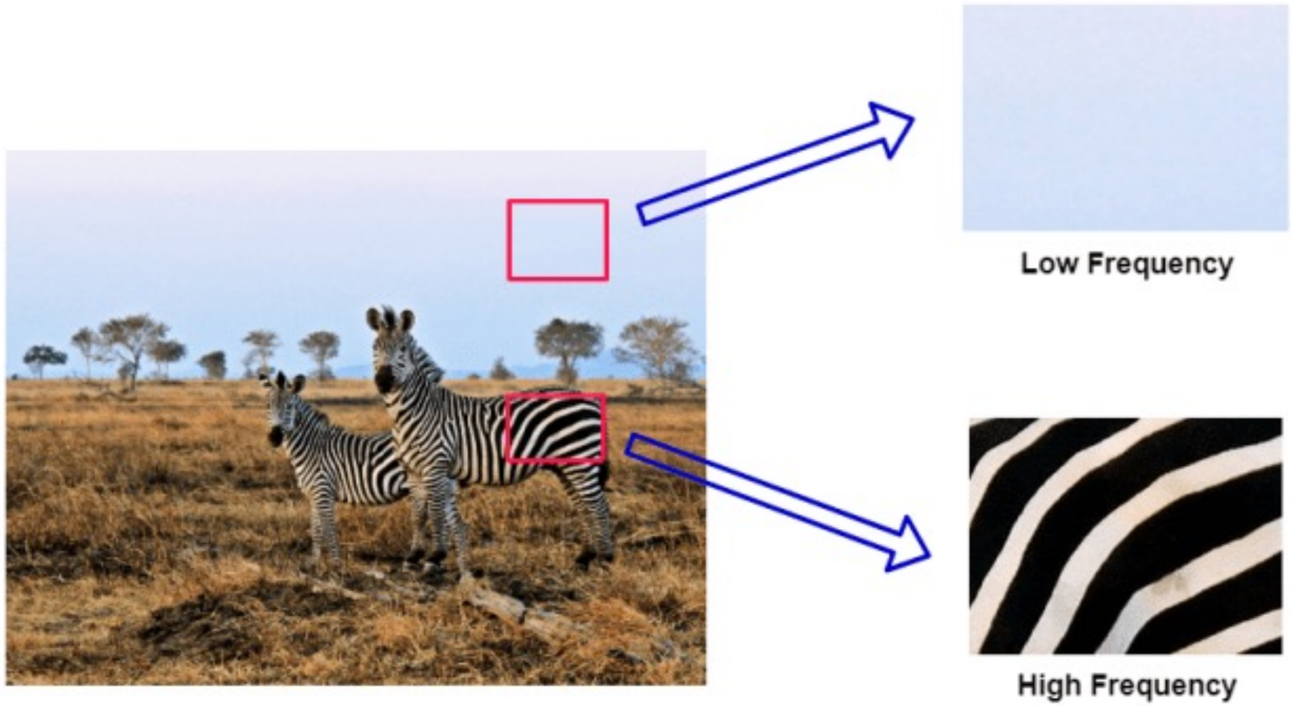

Image Filtering in the Frequency Domain

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- **LOW frequency = smooth regions**
  - Sky
  - Large uniform areas
  - Slowly varying intensity
- **HIGH frequency = rapid changes**
  - Edges
  - Fine textures
  - Noise
- **Understanding sharpening and smoothing**
  - Smoothing = removing high frequencies
  - Sharpening = amplifying high frequencies

Image Filtering in the Frequency Domain

From Spatial to Frequency Domain

- **Spatial domain:**
  We analyze pixel values directly.
- **Frequency domain:**
  We analyze how pixel values change across space.

    Low frequencies → smooth variations

    High frequencies → edges and fine details

Image Filtering in the Frequency Domain

The key idea:

Images can be decomposed into sinusoids

Any image can be represented as a sum of simple wave patterns of different frequencies.

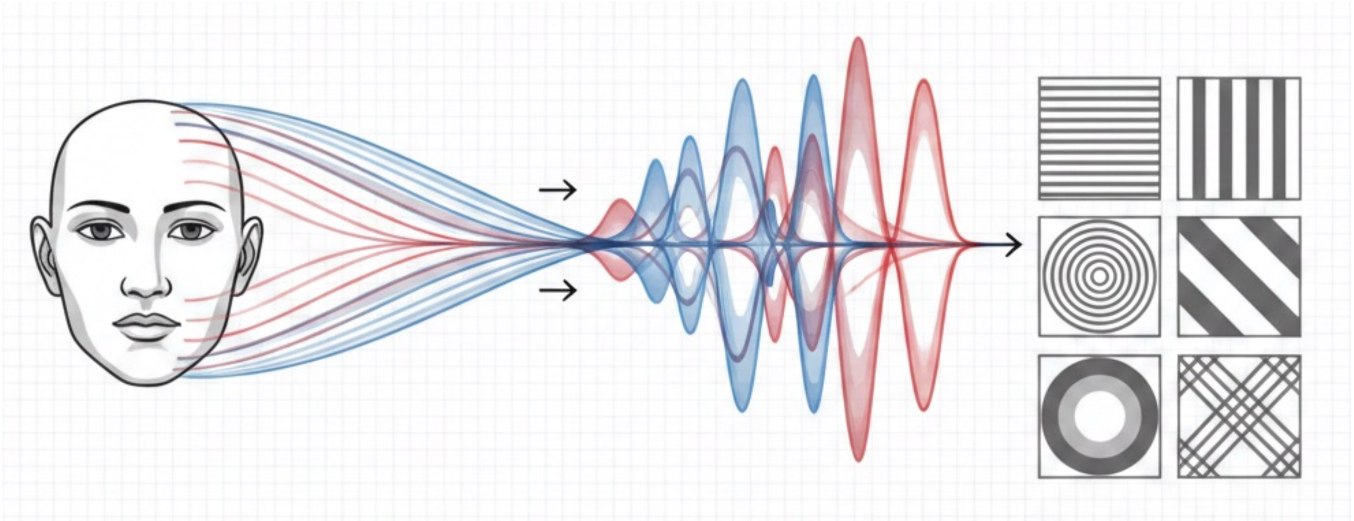


Image Filtering in the Frequency Domain

Who is Fourier?

Jean Baptiste Joseph Fourier

French mathematician and physicist

21 March 1768 - 17 May 1830.

He was actually obsessed with heat.

He was very interested in knowing how heat propagated through materials of different shapes.

=> That is what led him to develop the Fourier transform.

Image Filtering in the Frequency Domain

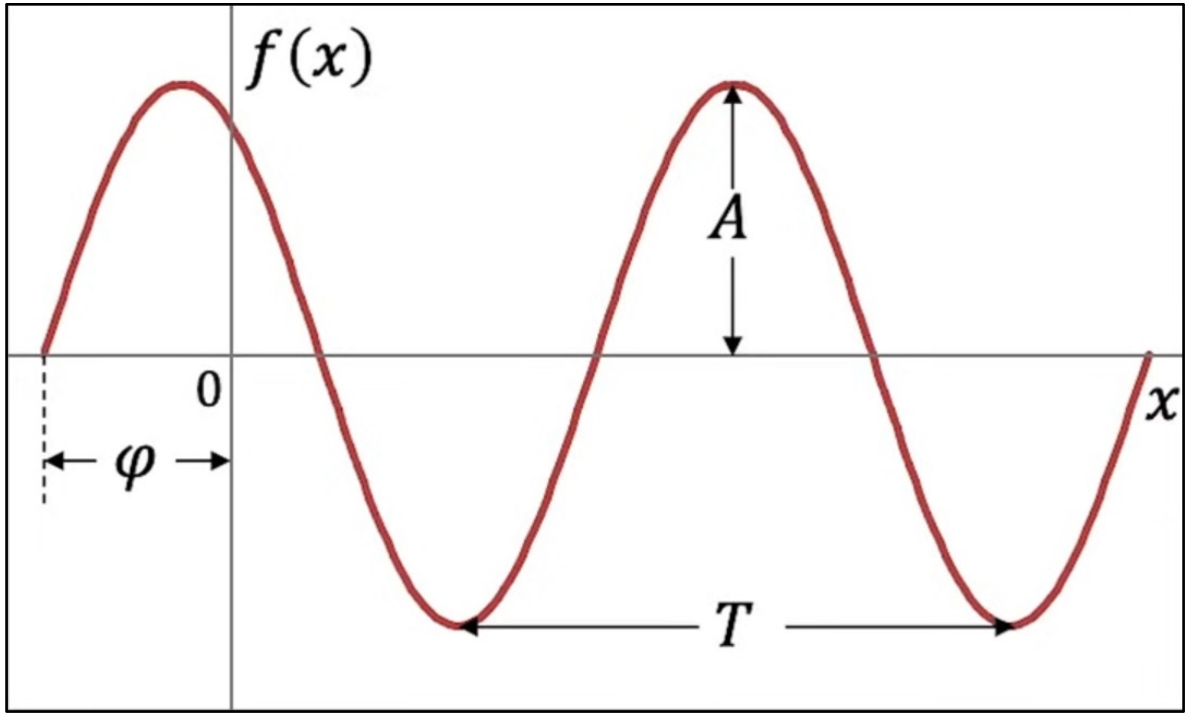
Fourier Transform

- Basically, any periodic function can actually be written as a weighted sum of infinite sinusoids of different frequencies.
- Functions that are not periodic (but whose area under the curve is finite) can also be expressed as the integral of sines and cosines multiplied by a weighting function.

Image Filtering in the Frequency Domain

Sinusoid

$$f(x) = A\,sin(2\,\pi\,u\,x + \varphi)$$



x: the time/position (in images)

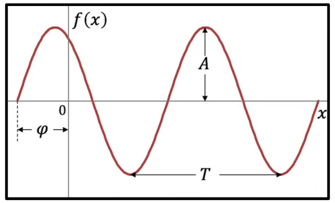
A: amplitude (the peak value of the wave)

T: period

φ: phase (a shift in time, measured in radians)

u: frequency (how many cycles per second, measured in Hz $= 1/T$)

Image Filtering in the Frequency Domain



Generating the Sine and Cosine Graphs from the Unit Circle

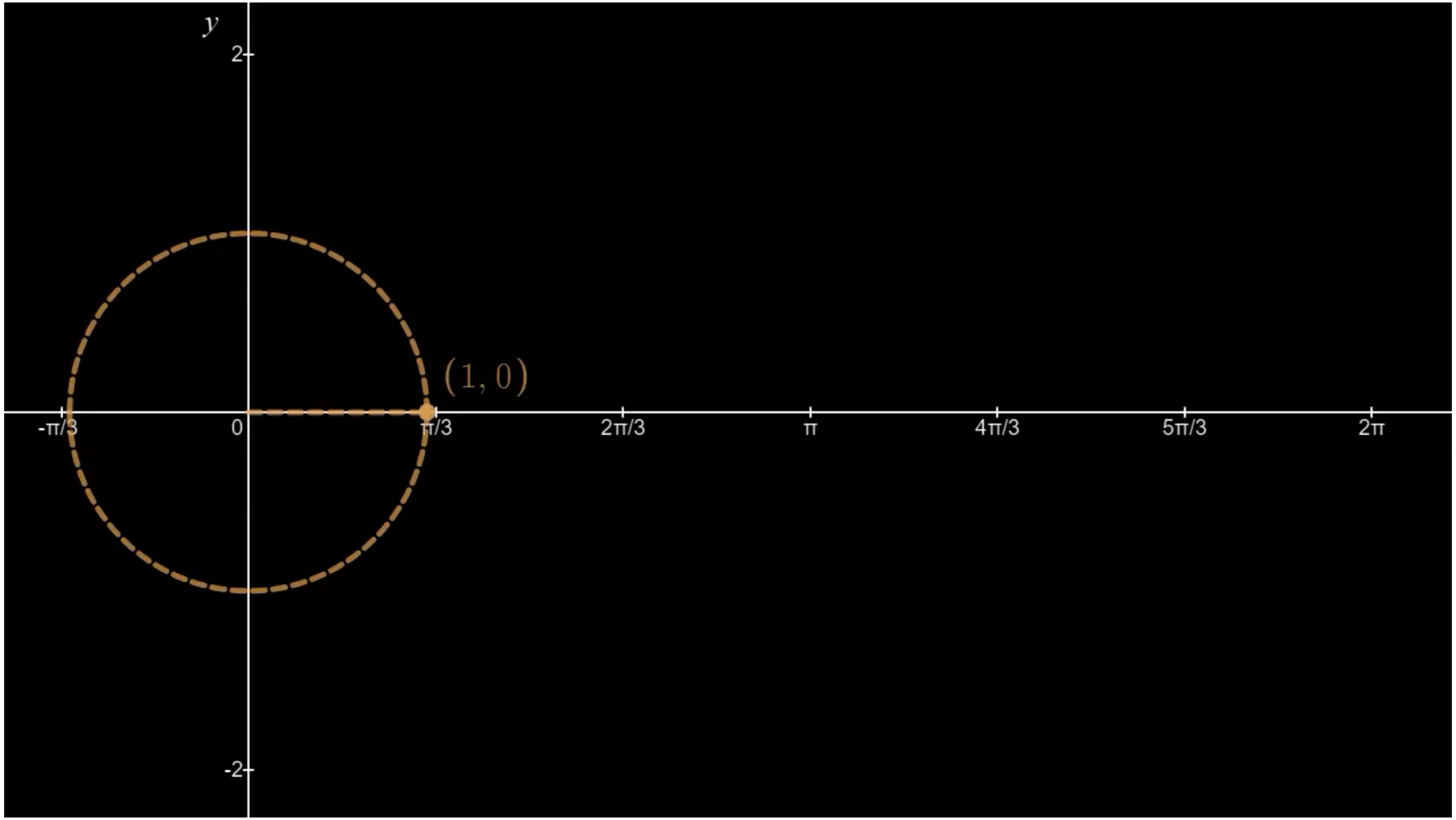


Image Filtering in the Frequency Domain

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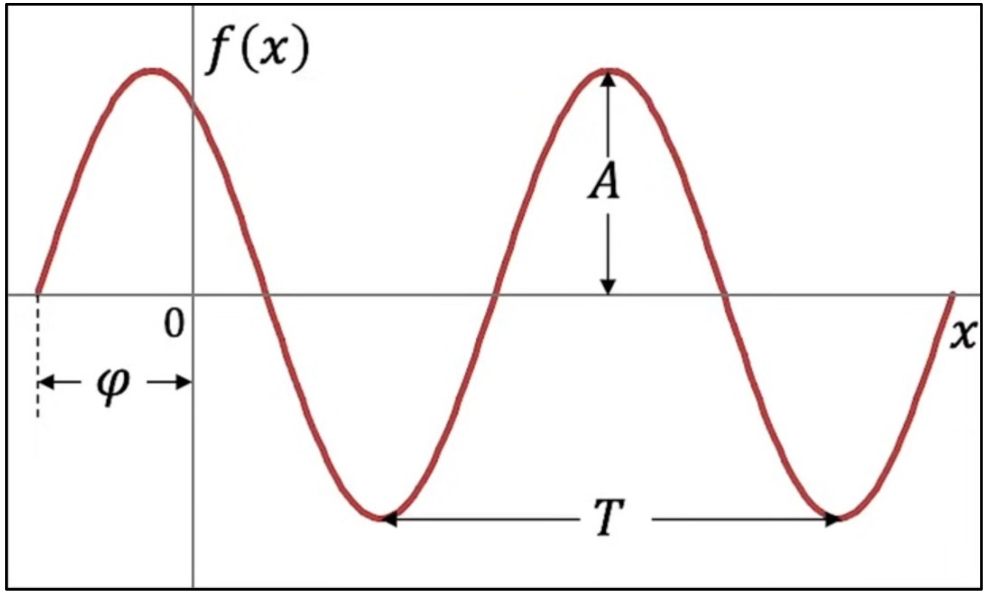
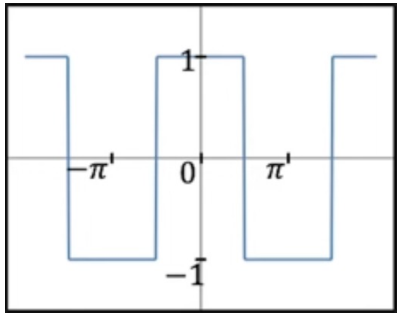


A sinusoid is defined by:

1. How strong it is (amplitude),
2. How fast it oscillates (frequency), and,
3. Where it starts (phase).

Image Filtering in the Frequency Domain

Fourier transform to decompose any periodic function



Square wave (period = 2 π).

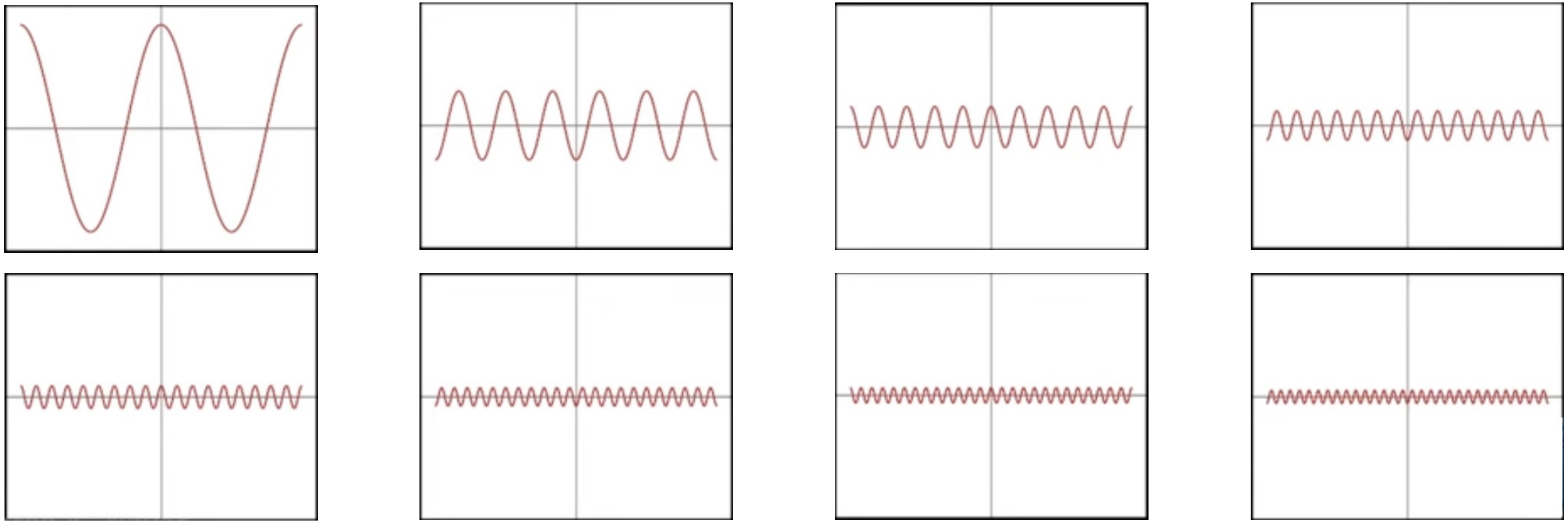
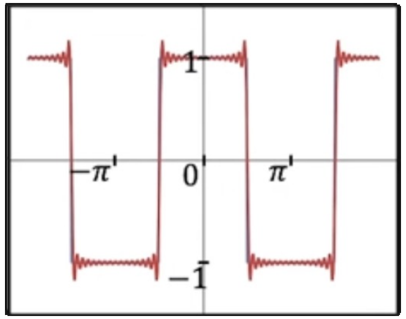


Image Filtering in the Frequency Domain

Fourier transform to decompose any periodic function



Square wave (period = 2 π).

Sum of sinusoids

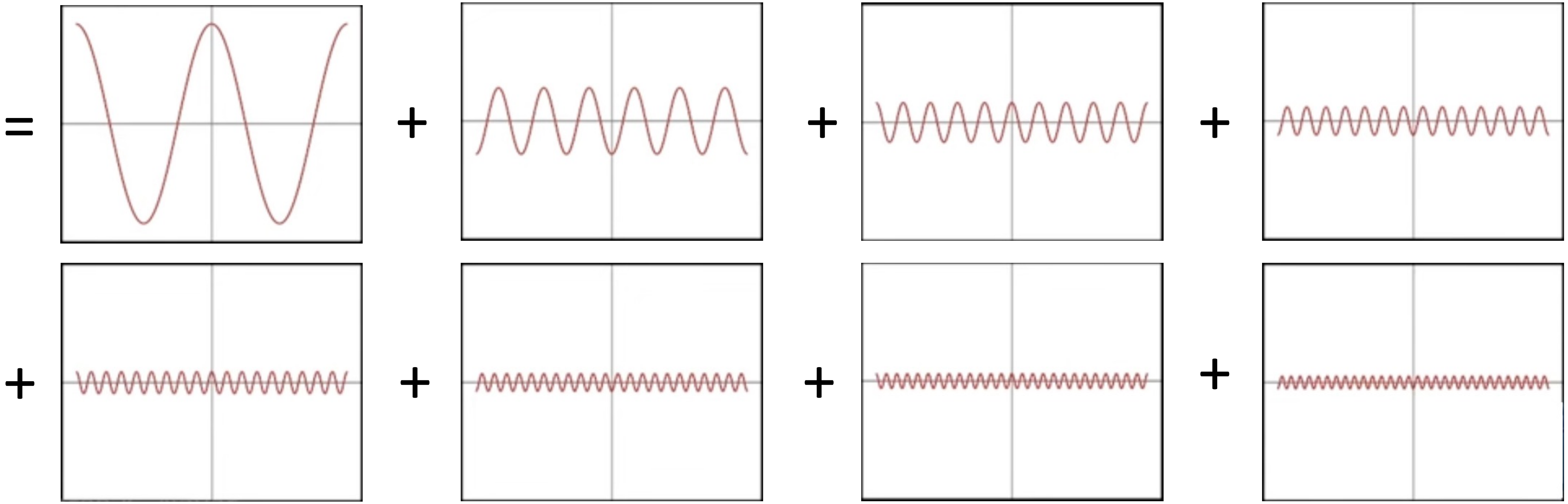
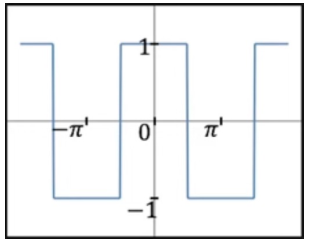


Image Filtering in the Frequency Domain

Fourier transform to decompose any periodic function



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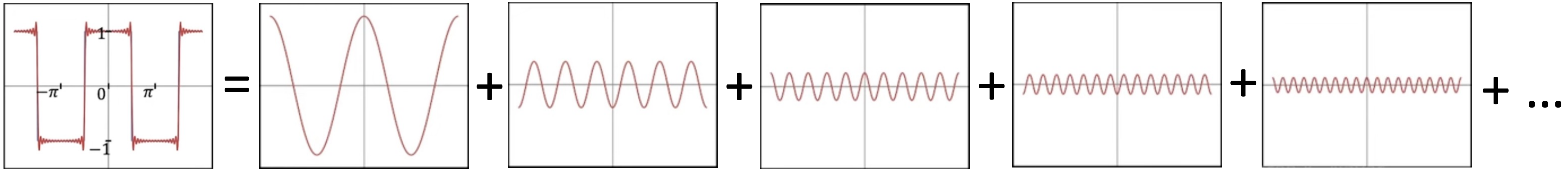


Sum of sinusoids

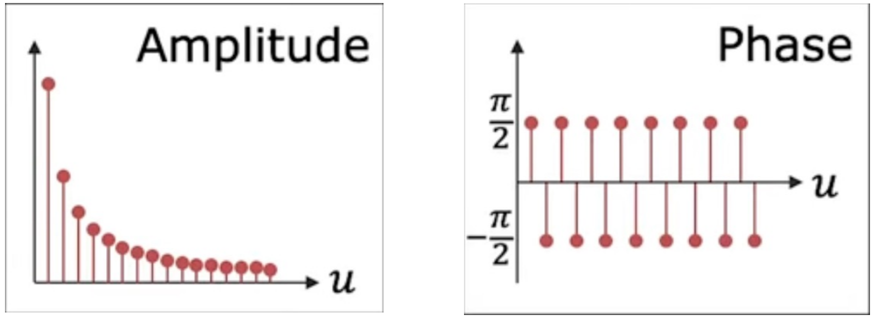


Image Filtering in the Frequency Domain

Fourier transform:

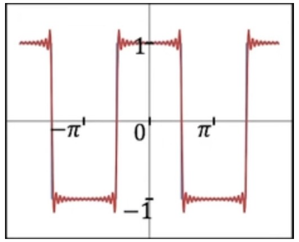
- It represents a signal ***f(x)*** in terms of **amplitudes** and **phases** of **its** constituent **sinusoids**.

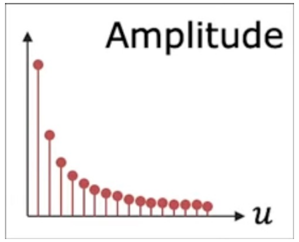
$$f(x) \quad \text{->} \quad \textbf{FT} \quad \text{->} \quad F(u)$$

Inverse Fourier transform:

- It computes the spatial signal ***f(x)*** from the **amplitudes** and **phases** of the constituent **sinusoids**.

$$F(u) \quad \text{->} \quad \textbf{IFT} \quad \text{->} \quad f(x)$$





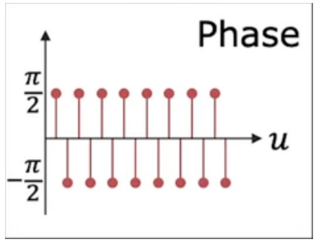


Image Filtering in the Frequency Domain

Fourier Transform (of a 1D signal):

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi ux} dx$$

Inverse Fourier Transform (of a 1D signal):

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{i2\pi ux} du$$

x: space

u: frequency

$e^{i\theta} = \cos\theta + i\sin\theta$

$i = \sqrt{-1}$

Image Filtering in the Frequency Domain

Why do we have $e^{i\theta} = \cos\theta + i\sin\theta$? $\quad (i = \sqrt{-1})$

Expand $e^{i\theta}$ using Taylor Series:

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \frac{(i\theta)^6}{6!} + \cdots$$

$$e^{i\theta} = \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots\right)}_{\cos\theta} + i\underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots\right)}_{\sin\theta}$$

Image Filtering in the Frequency Domain

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi ux} dx$$

- **Conceptually:** Fourier coefficients are not just numbers.

  They encode amplitude & phase.

- **Mathematically:** The transform produces **complex numbers**.

Image Filtering in the Frequency Domain

$\Rightarrow$ **The Fourier Transform is Complex!**

$F(u)$ holds the Amplitude and the Phase of the sinusoid of frequency u

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi ux} dx$$

$$F(u) = \Re\{F(u)\} + i\,\Im\{F(u)\} = |A(u)|\, e^{i\varphi(u)}$$

Amplitude: $A(u) = \sqrt{\Re\{F(u)\}^2 + \Im\{F(u)\}^2}$

Phase: $\varphi(u) = \operatorname{atan2}(\Im\{F(u)\},\ \Re\{F(u)\})$

Image Filtering in the Frequency Domain

Phase: $\varphi(u) = \text{atan2}(\Im\{F(u)\},\ \Re\{F(u)\})$

Why atan2?

For example:

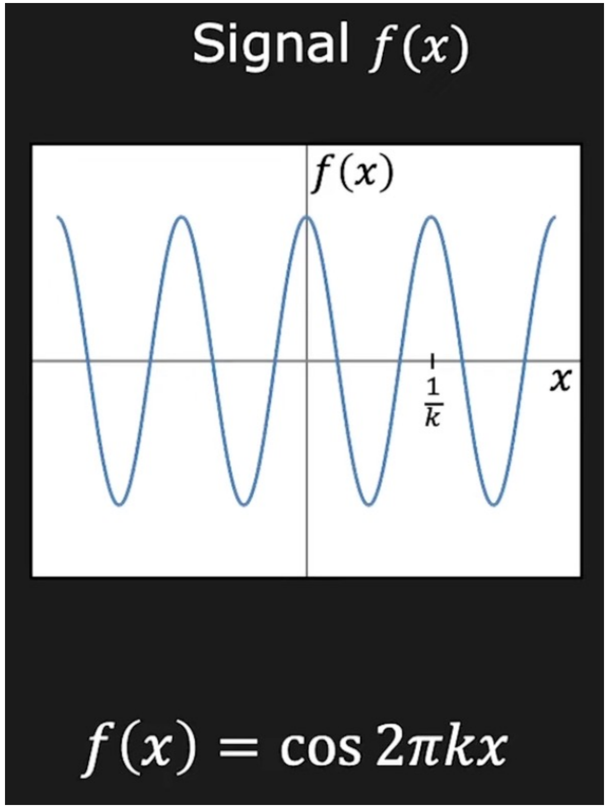
- $x = 1$, $y = 1$ → angle ≈ 45°
- $x = -1$, $y = -1$ → same ratio → 1

- (1,1) → 45°
- (-1,-1) → 225°

$$\text{atan2}(y, x) = \begin{cases} \arctan(y/x) & x > 0 \\ \arctan(y/x) + \pi & x < 0, y \geq 0 \\ \arctan(y/x) - \pi & x < 0, y < 0 \end{cases}$$

Image Filtering in the Frequency Domain

Few examples of the Fourier transform:



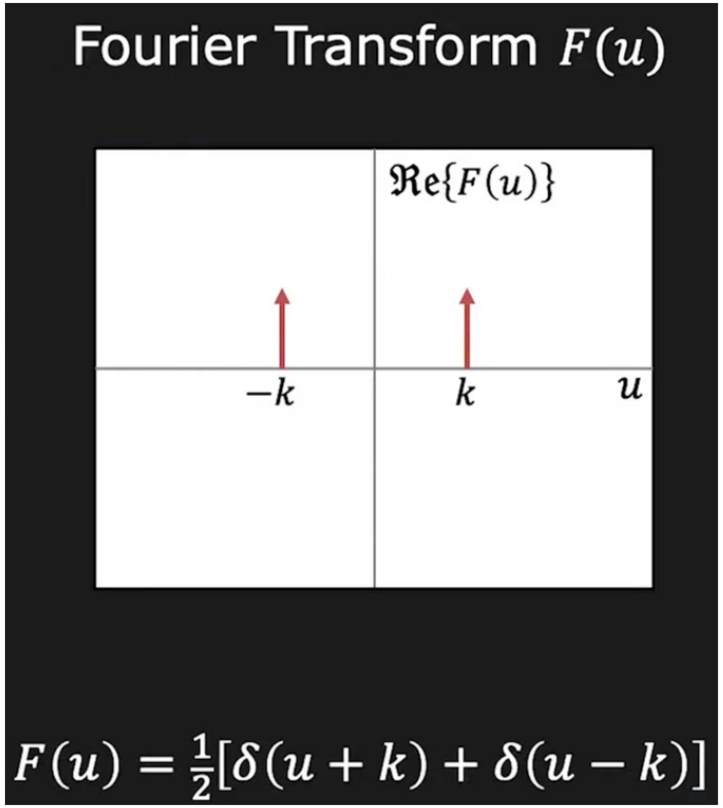
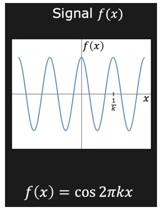
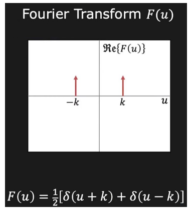


Image Filtering in the Frequency Domain





- We have: $e^{i\theta} = \cos\theta + i\sin\theta$
- and $e^{-i\theta} = \cos\theta - i\sin\theta$
- $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ => $\cos\theta = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right)$

$$\cos(2\pi kx) = \frac{1}{2}\left(e^{i2\pi kx} + e^{-i2\pi kx}\right)$$

So a cosine is actually:

- one positive frequency component $+k$
- one negative frequency component $-k$

Image Filtering in the Frequency Domain

- About negative frequencies (https://www.youtube.com/watch?v=SvFfVtRT1UI):

Image Filtering in the Frequency Domain

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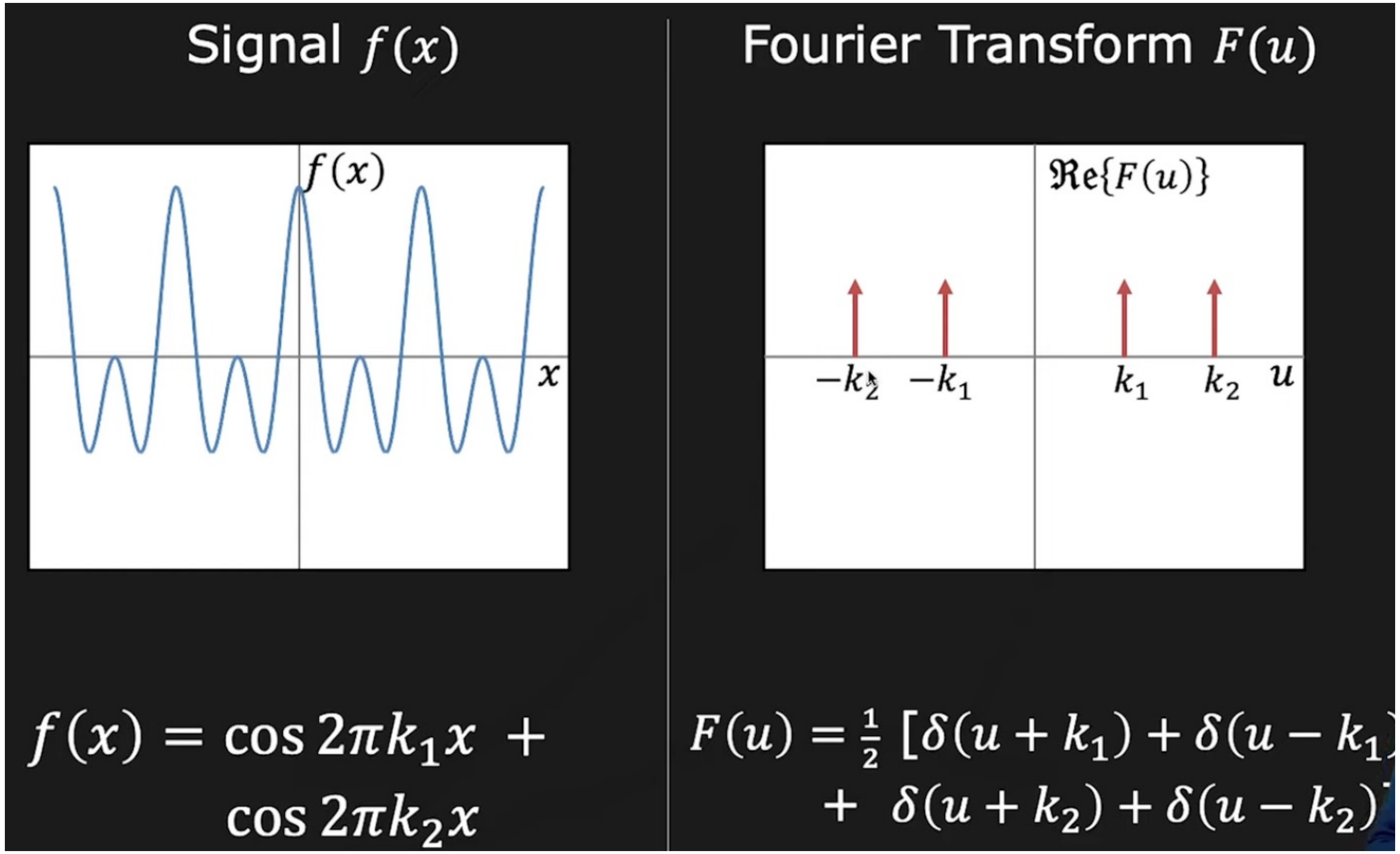

Image Filtering in the Frequency Domain

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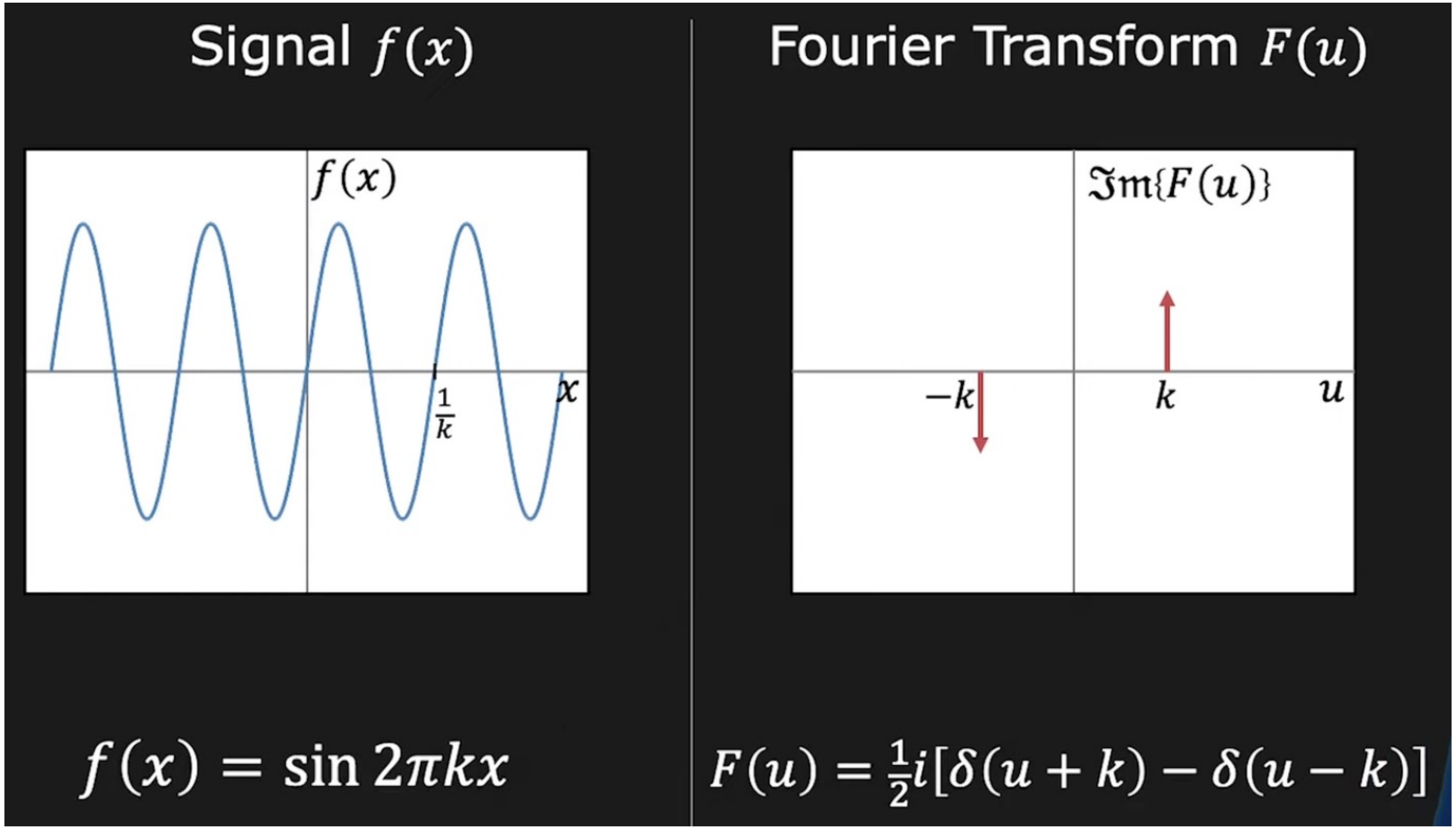

Image Filtering in the Frequency Domain

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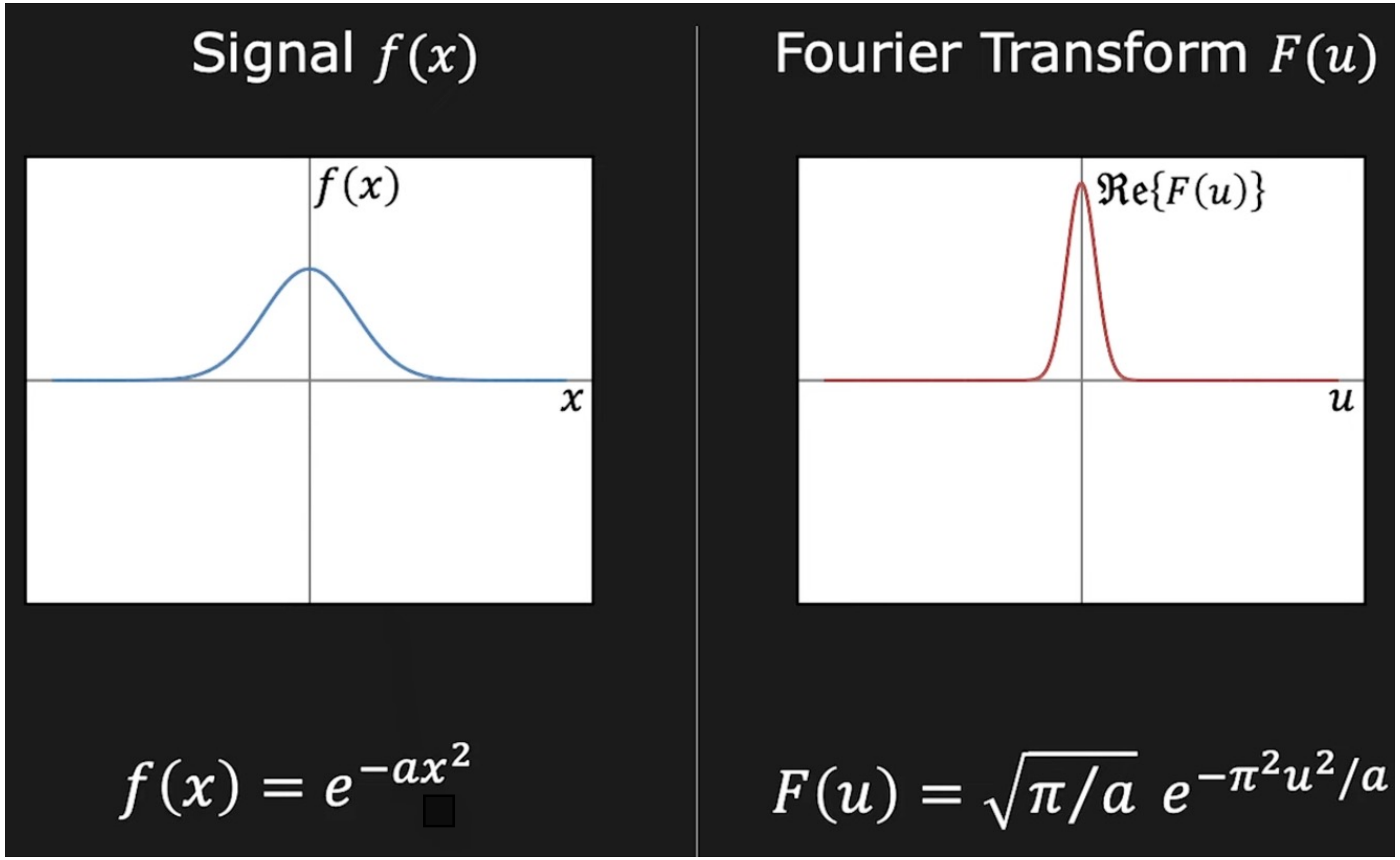


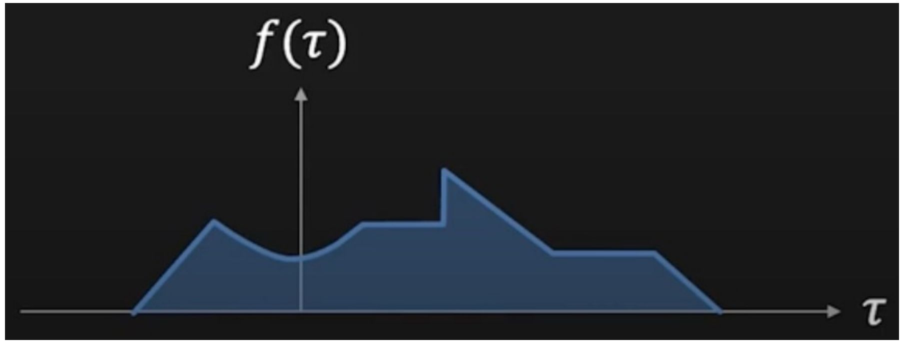
Image Filtering in the Frequency Domain

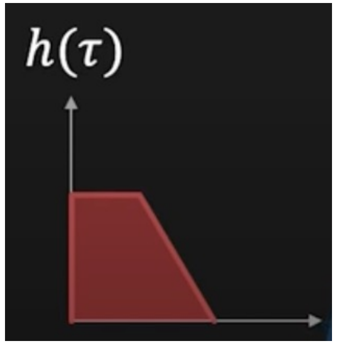
Fourier transform and convolution

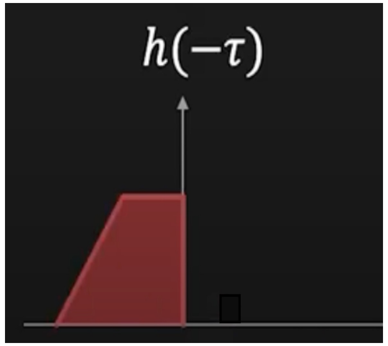
- A close and important relationship.

Convolution of 2 functions: *f(x)* and *h(x)*

$$g(x) = f(x) * h(x) = \int_{-\infty}^{\infty} f(\tau)\, h(x-\tau)\; d\tau$$







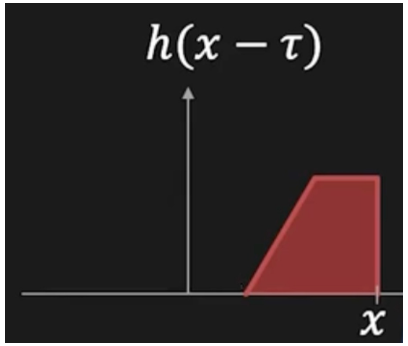


Image Filtering in the Frequency Domain

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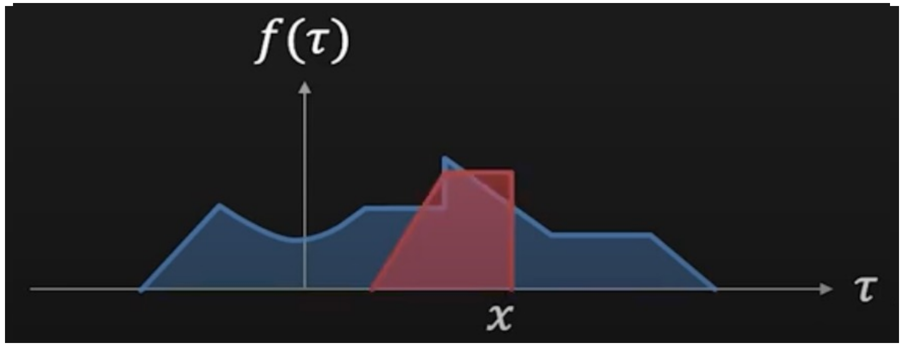

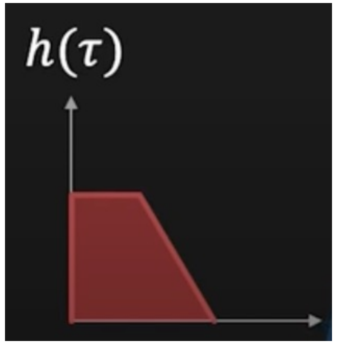

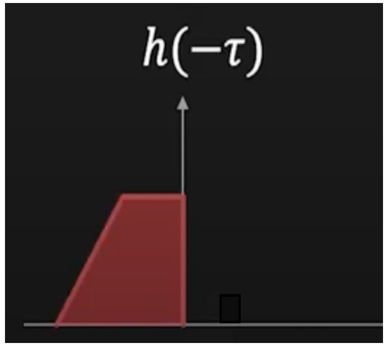

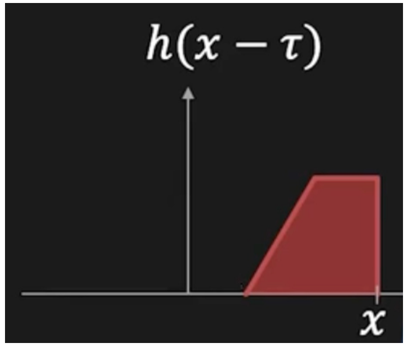

Image Filtering in the Frequency Domain

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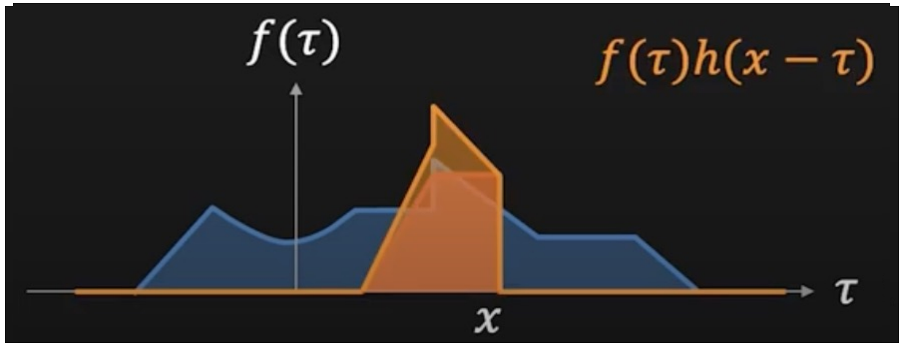

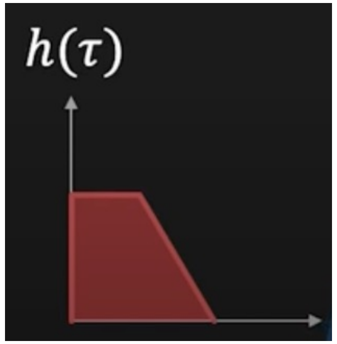

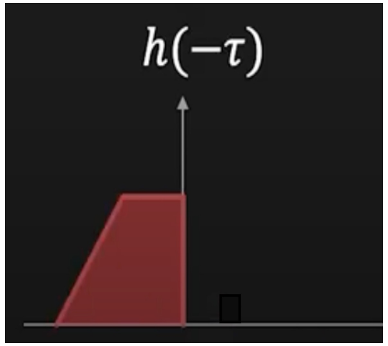

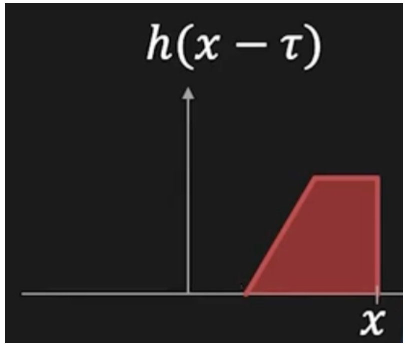

Image Filtering in the Frequency Domain

Fourier transform and convolution

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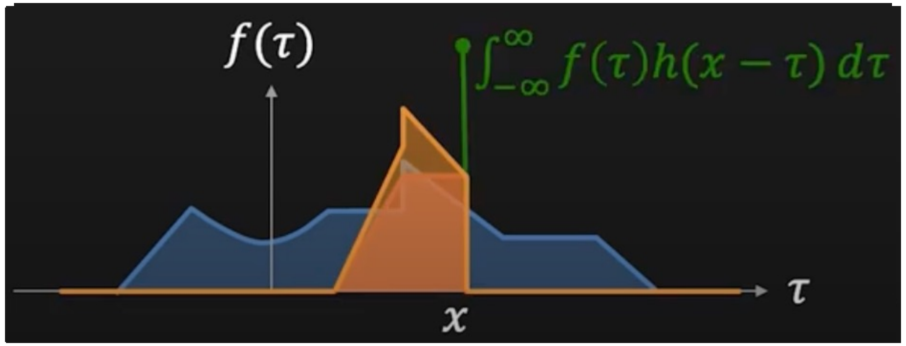

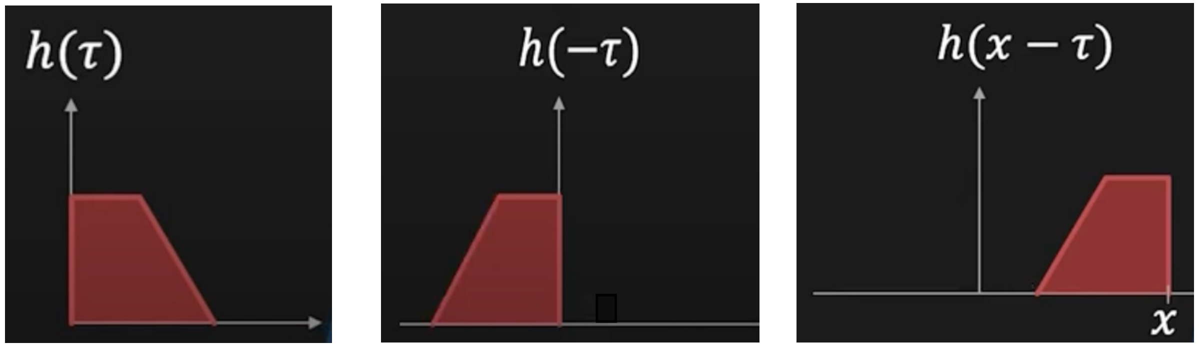

Image Filtering in the Frequency Domain

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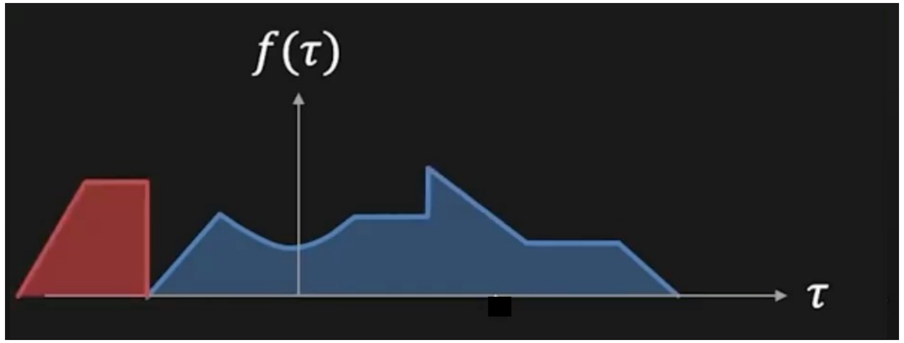
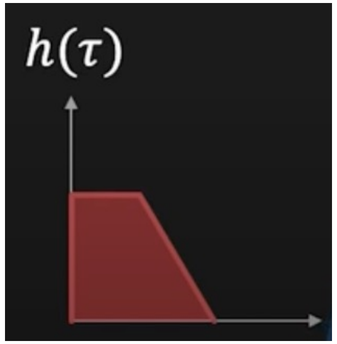
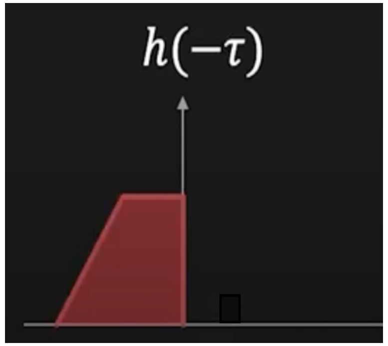






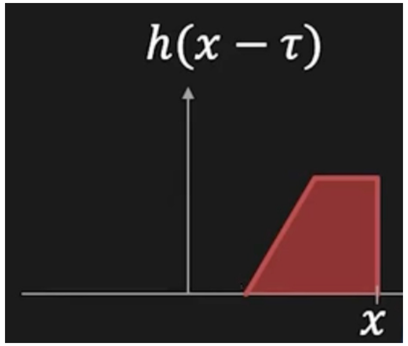
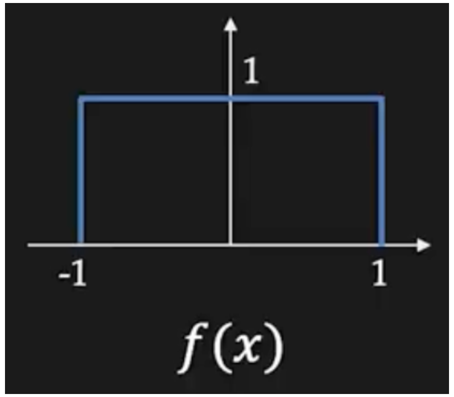
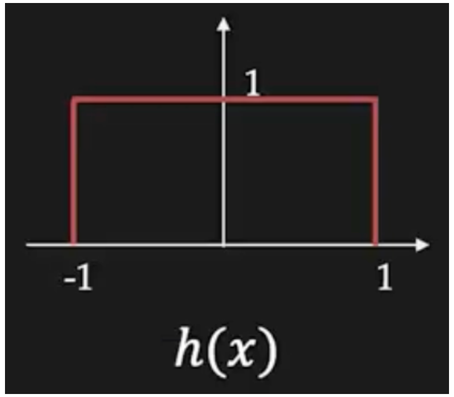


Image Filtering in the Frequency Domain

Fourier transform and convolution: Few examples





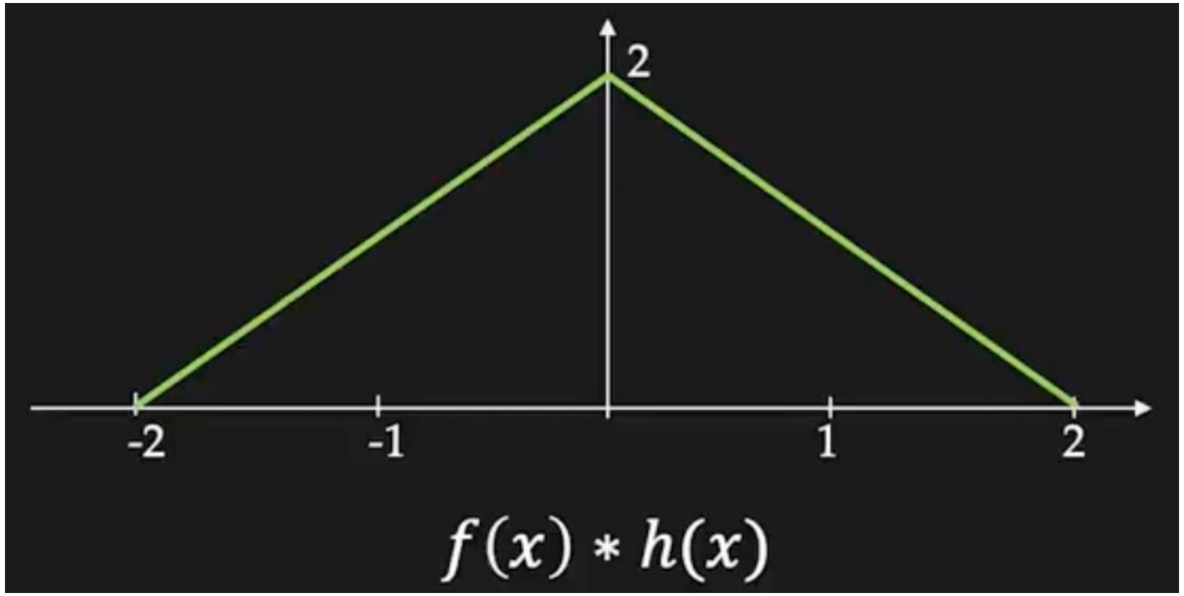


Image Filtering in the Frequency Domain

Fourier transform and convolution

$\Rightarrow$ The convolution theorem

Convolution: $g(x) = f(x) * h(x) = \int_{-\infty}^{\infty} f(\tau)\, h(x-\tau)\; d\tau$

Fourier Transform of $g(x)$:

$$G(u) = \int_{-\infty}^{\infty} g(x)\, e^{-i2\pi ux}\; dx = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\tau)\, h(x-\tau)\, e^{-i2\pi ux}\, d\tau\, dx$$

$$G(u) = \underbrace{\int_{-\infty}^{\infty} f(\tau)\, e^{-i2\pi u\tau}\; d\tau}_{F(u)}\; \underbrace{\int_{-\infty}^{\infty} h(x-\tau)\, e^{-i2\pi u(x-\tau)}\, dx}_{H(u)} = F(u)\, H(u)$$

Image Filtering in the Frequency Domain

Fourier transform and convolution

Spatial Domain		Frequency Domain
$g(x) = f(x) * h(x)$ Convolution	$\longleftrightarrow$	$G(u) = F(u)\,H(u)$ Multiplication
$g(x) = f(x)\,h(x)$ Multiplication	$\longleftrightarrow$	$G(u) = F(u) * H(u)$ Convolution

Image Filtering in the Frequency Domain

Fourier transform and convolution

- Very efficient implementation of FT and IFT are available.
- Working in the frequency domain is more efficient.
- Analyze in the frequency domain, a filter that is designed in the spatial domain.

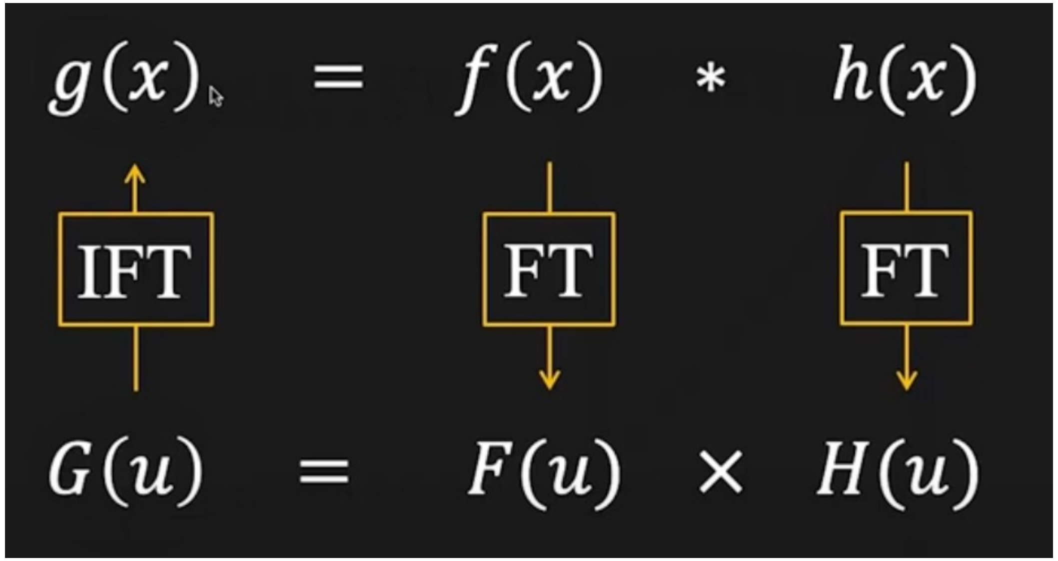
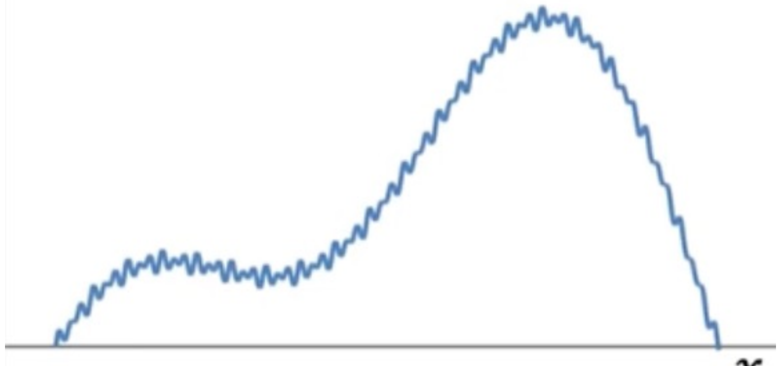
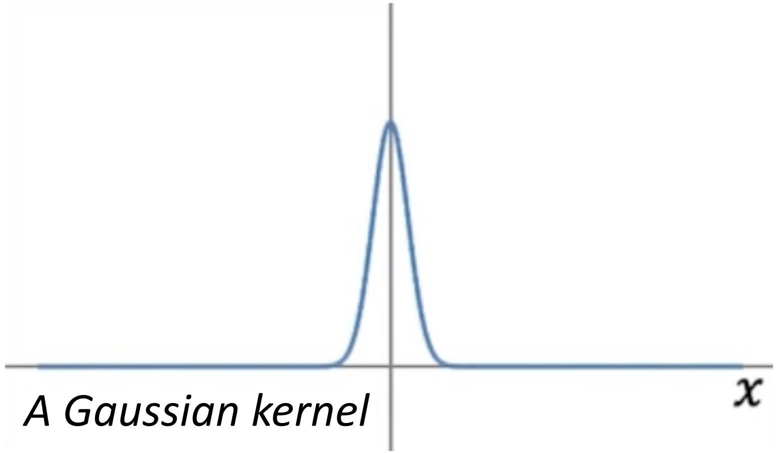


Image Filtering in the Frequency Domain

Fourier transform and convolution



A noisy signal f(x)



A Gaussian kernel

⇒ Convolving the noisy signal with the Gaussian kernel.

Image Filtering in the Frequency Domain

Fourier transform and convolution

⇒ Convolving the noisy signal with the Gaussian kernel.

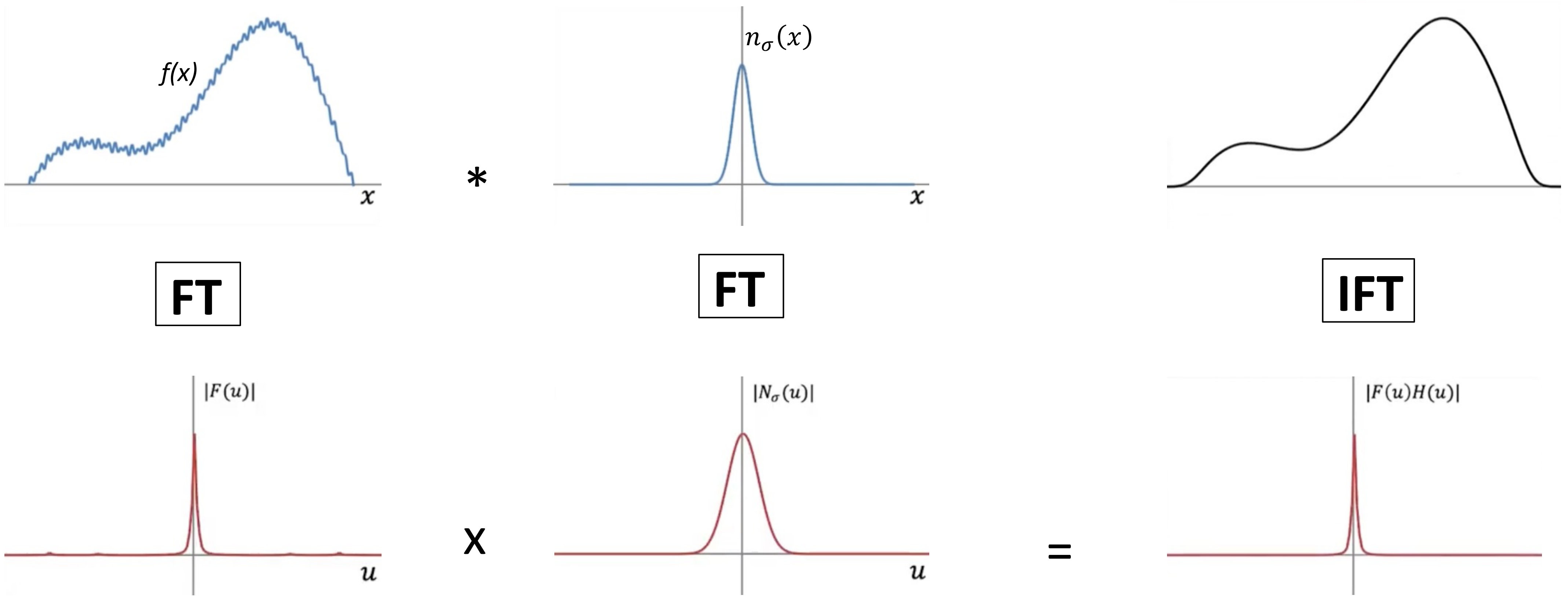


Image Filtering in the Frequency Domain

Application to images – The 2D Fourier transform

Fourier Transform: $F(u,v) = \iint_{-\infty}^{\infty} f(x,y)\, e^{-i2\pi(ux+vy)} dx\, dy$

u and *v* are the frequencies along x and y, respectively.

Inverse Fourier Transform: $f(x,y) = \iint_{-\infty}^{\infty} F(u,v)\, e^{i2\pi(ux+vy)} du\, dv$

Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

Discrete Fourier Transform:

$$F[p,q] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f[m,n] e^{-i2\pi pm/M} e^{-i2\pi qn/N}$$

$p = 0 \ldots M-1$ and $q = 0 \ldots N-1,$

p and *q* are the frequencies along m and n, respectively.

Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

Inverse Discrete Fourier Transform:

$$f[m,n] = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} F[p,q] e^{i2\pi pm/M} e^{i2\pi qn/N}$$

$m = 0 \ldots M-1$ and $n = 0 \ldots N-1$

Image Filtering in the Frequency Domain

The 2D Discrete Fourier transform (DFT)

$$F[p,q] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f[m,n] e^{-i2\pi pm/M} e^{-i2\pi qn/N}$$

$$f[m,n] = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} F[p,q] e^{i2\pi pm/M} e^{i2\pi qn/N}$$

Formulas used in MATLAB / NumPy FFT

The Fourier transform adds contributions from all MN pixels, the resulting coefficients are MN times larger.

The inverse transform divides by MN so that the original image is correctly reconstructed.

Image Filtering in the Frequency Domain

The 2D Discrete Fourier transform (DFT): Other ways

1. Normalization in the inverse transform

$$F = \mathrm{DFT}(f)$$

$$f = \frac{1}{MN}\,\mathrm{IDFT}(F)$$

2. Normalization in the forward transform

$$F = \frac{1}{MN}\,\mathrm{DFT}(f)$$

$$f = \mathrm{IDFT}(F)$$

3. Symmetric normalization

$$F = \frac{1}{\sqrt{MN}}\,\mathrm{DFT}(f)$$

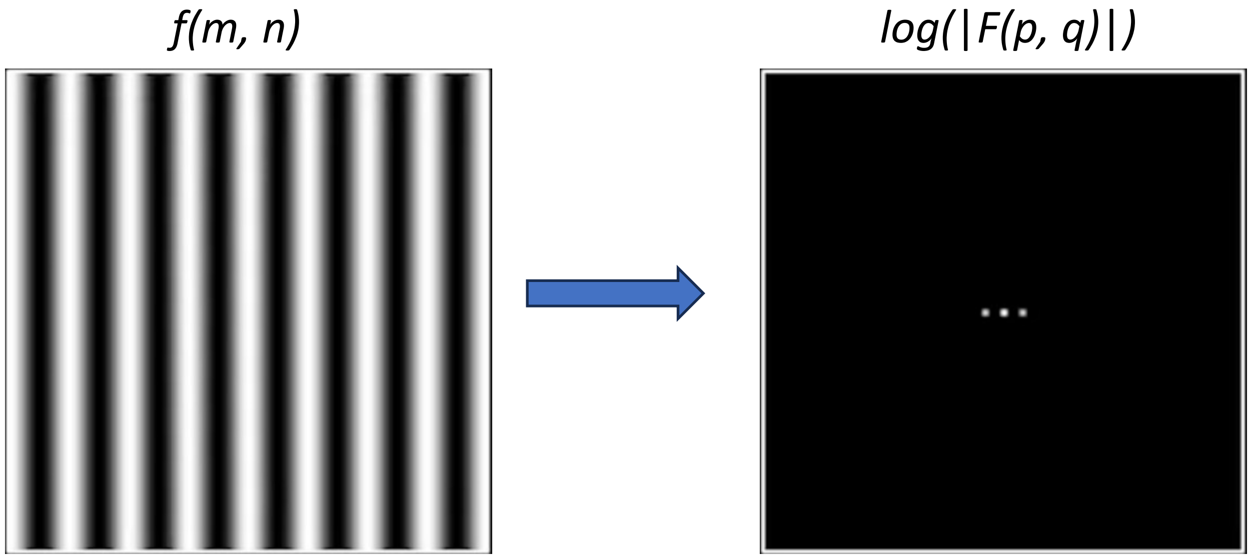
$$f = \frac{1}{\sqrt{MN}}\,\mathrm{IDFT}(F)$$

These three conventions are mathematically equivalent.

They only differ in where the normalization factor is applied so that the forward and inverse transforms remain consistent.

Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)



Why are the points on the horizontal axis?

The image intensity varies only along x.

p → frequency along x
q → frequency along y

Since the image does not vary vertically, we have:

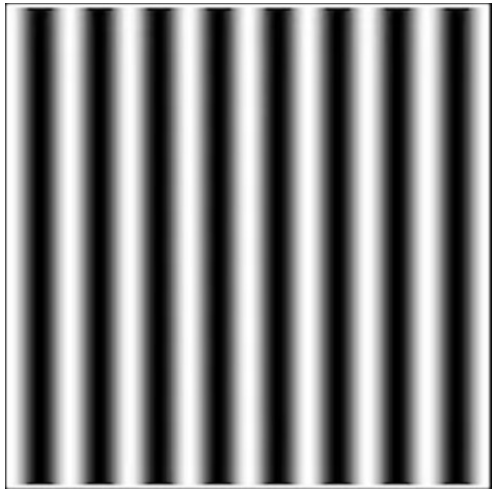
$$q = 0$$

⇒ All the spectral energy lies on the horizontal frequency axis.

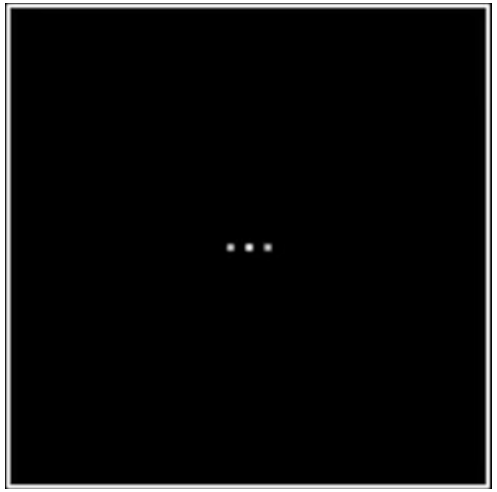
Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

f(m, n)



log(|F(p, q)|)



What the image *f* represents?

f contains vertical sinusoidal stripes.

=> The intensity varies periodically along the horizontal direction.

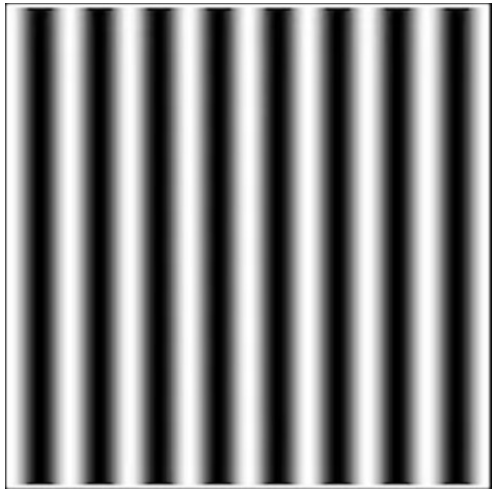
Mathematically, this image is close to:

$$f(x, y) = cos(2\pi u_0 x)$$

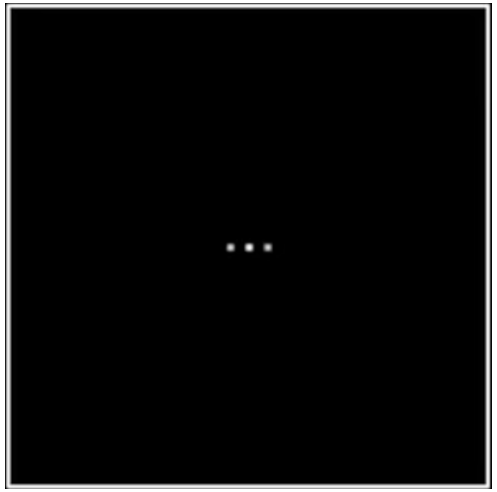
Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

f(m, n)



log(|F(p, q)|)



Why do we see three bright points?

The two symmetric points correspond to the frequencies:

$$(+u_0, 0) \text{ and} (-u_0, 0)$$

The center point corresponds to:

$$(p, q) = (0, 0)$$

This coefficient corresponds to the average brightness of the image.

$\Rightarrow$ **It is the DC coefficient.**

Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

The DC coefficient.

$$F[p,q] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f[m,n] e^{-i2\pi pm/M} e^{-i2\pi qn/N} \qquad F[0,0] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f[m,n]$$

$$f(x, y) = \bar{f} + variations$$

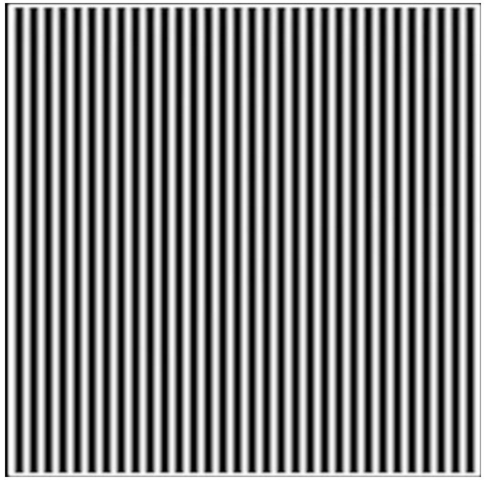
$\bar{f}$: average intensity (DC component)

variations: Other frequencies

Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

f(m, n)



log(|F(p, q)|)

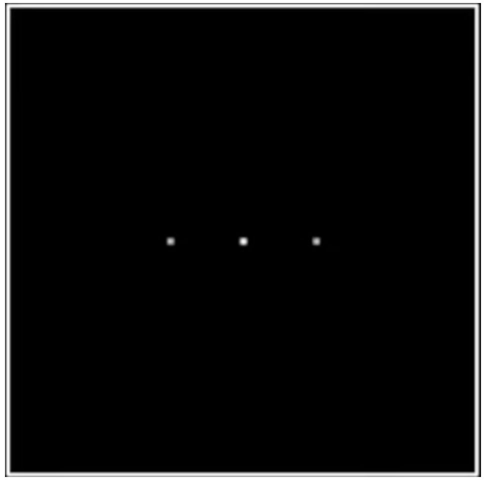
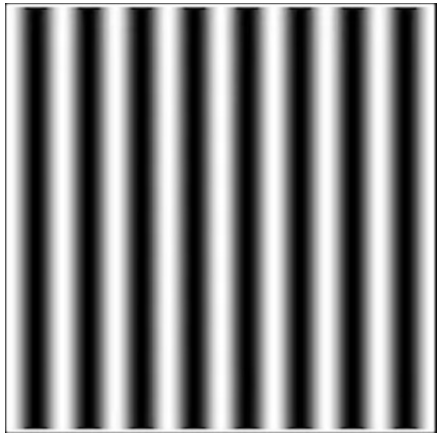
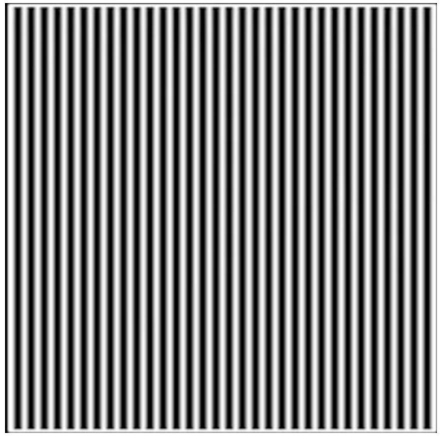


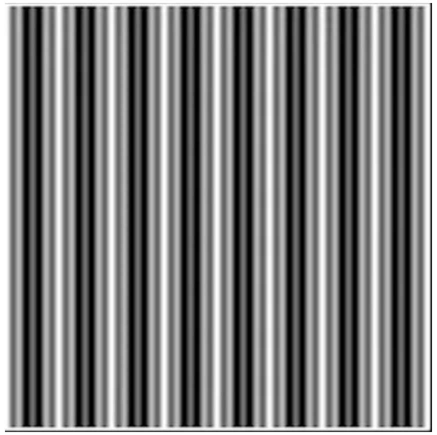
Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)





f(m, n)



log(|F(p, q)|)

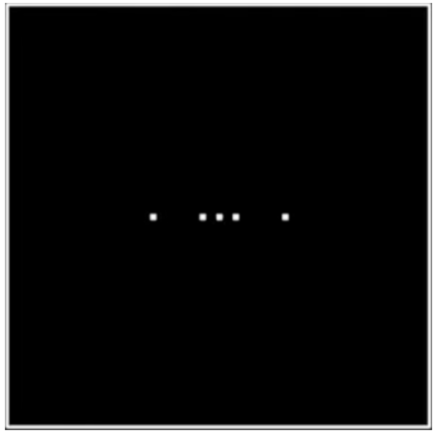


Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

The Fourier spectrum encodes orientation

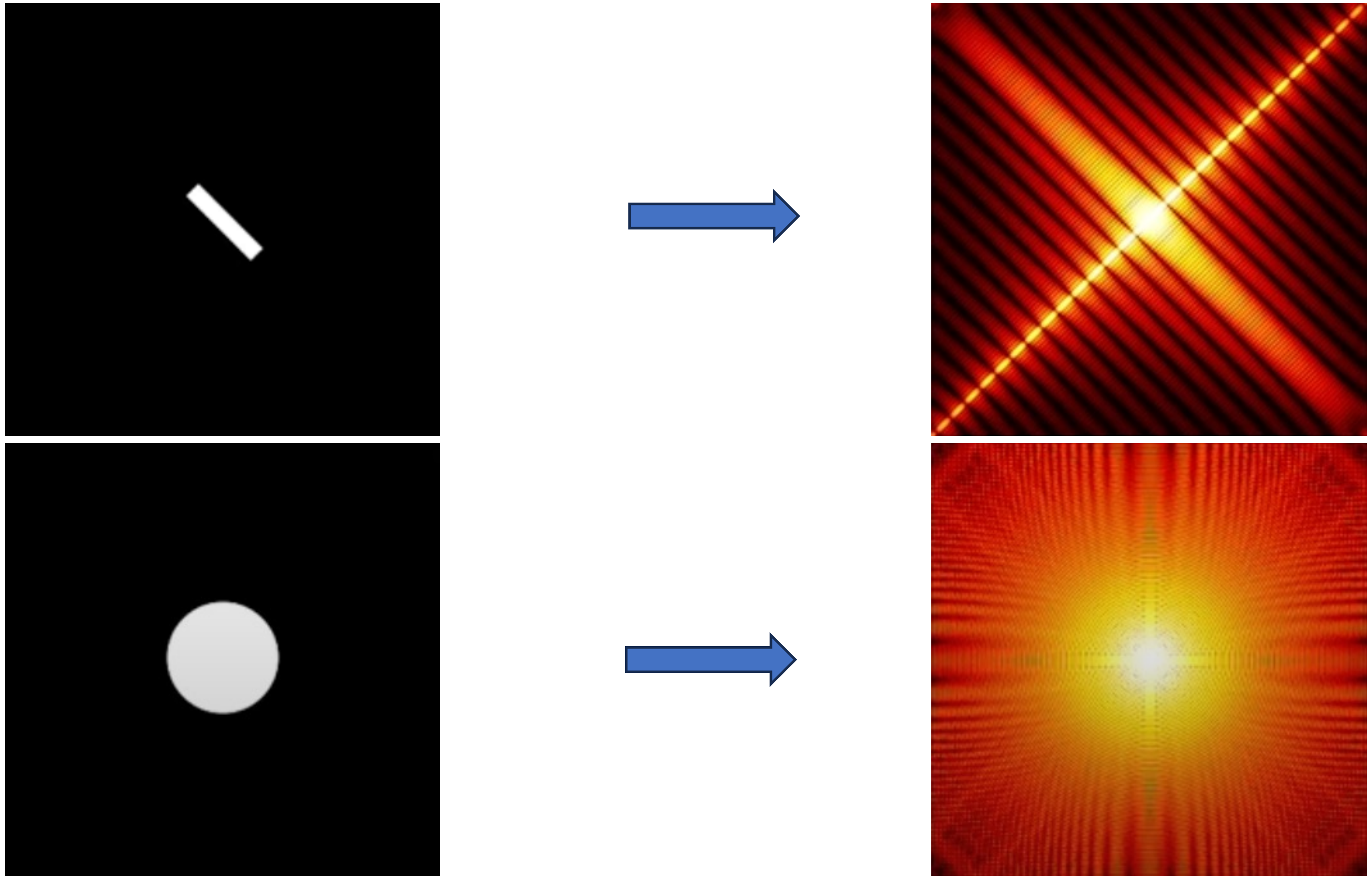
Image	**Spectrum**
Vertical stripes	Horizontal points
Horizontal stripes	Vertical points
Diagonal stripes	Diagonal points

$\Rightarrow$ **Orientation in image $\perp$ orientation in Fourier domain**

Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

f(m, n) *log(|F(p, q)|)*



Min Max

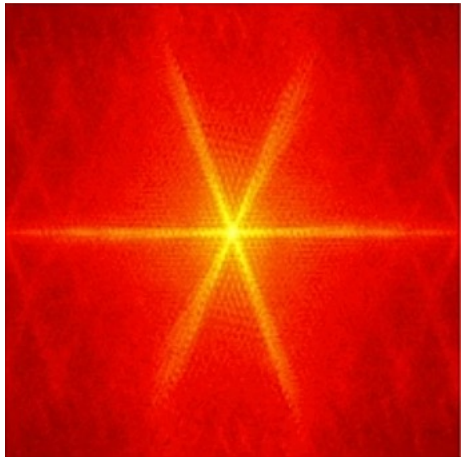
Color to show the magnitude

Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

f(m, n)

log(|F(p, q)|)



Min Max

Color to show the magnitude

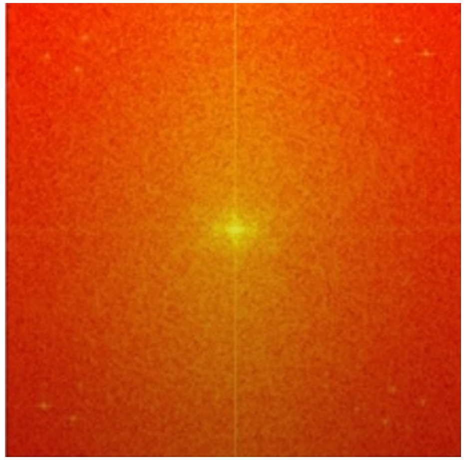
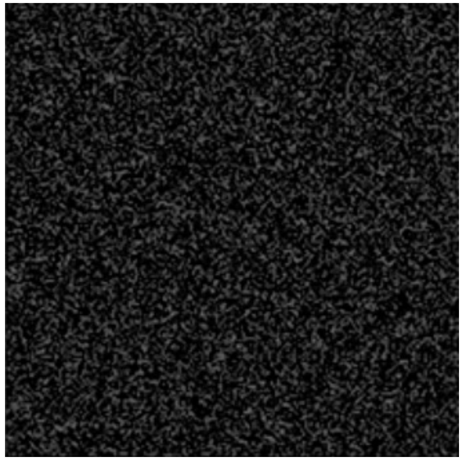


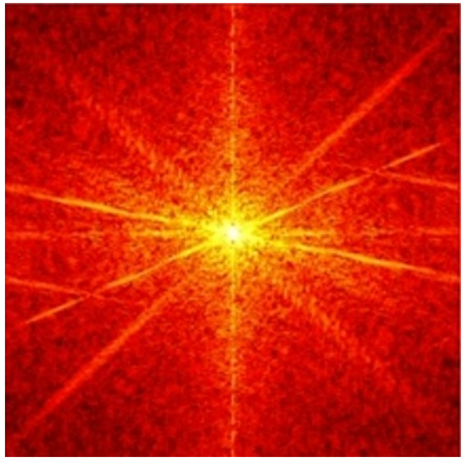
Image Filtering in the Frequency Domain

Application to images – The 2D Discrete Fourier transform (DFT)

f(m, n)



log(|F(p, q)|)



Color to show the magnitude

Image Filtering in the Frequency Domain

Application to image filtering: The low pass filter and the high pass filter

- Low frequencies in the transform are related to slowly varying intensity components.
- High frequencies are caused by sharp transitions in intensity, such as edges, corners and noise

$\Rightarrow$ A filter ***H(u, v)*** that attenuates high frequencies while passing low frequencies (low-pass filter) blurs an image

$\Rightarrow$ A high-pass filter (which attenuates low frequencies) enhances sharp detail, but cause a reduction in contrast in the image.

Image Filtering in the Frequency Domain

Application to image filtering: The low pass filter

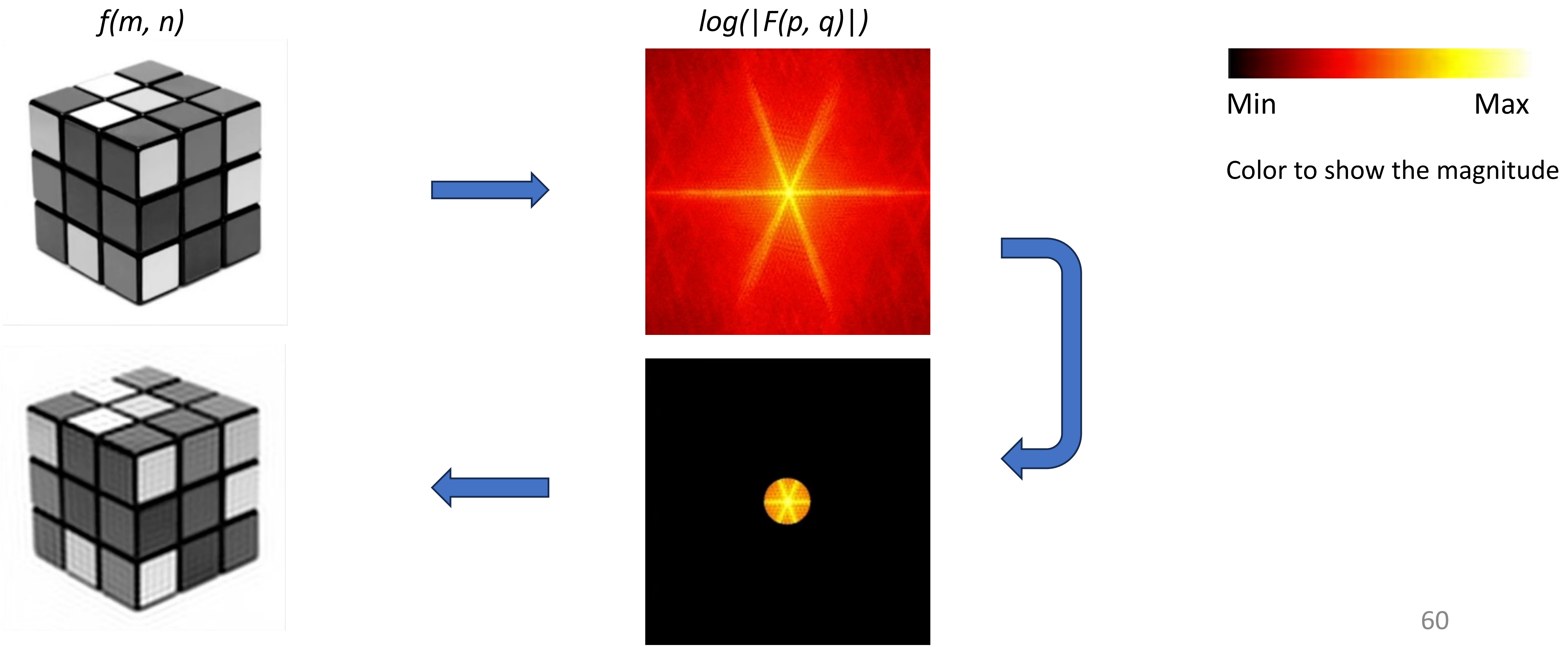
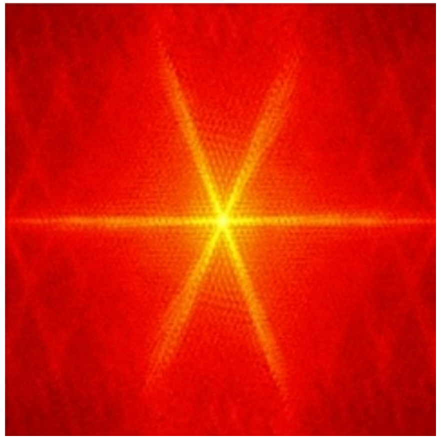


Image Filtering in the Frequency Domain

Application to image filtering: The low pass filter

$f(m, n)$

$\log(|F(p, q)|)$



Min Max

Color to show the magnitude

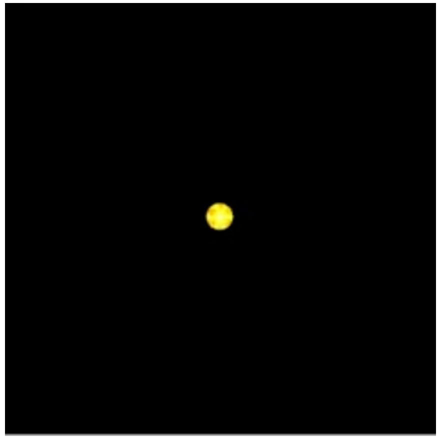
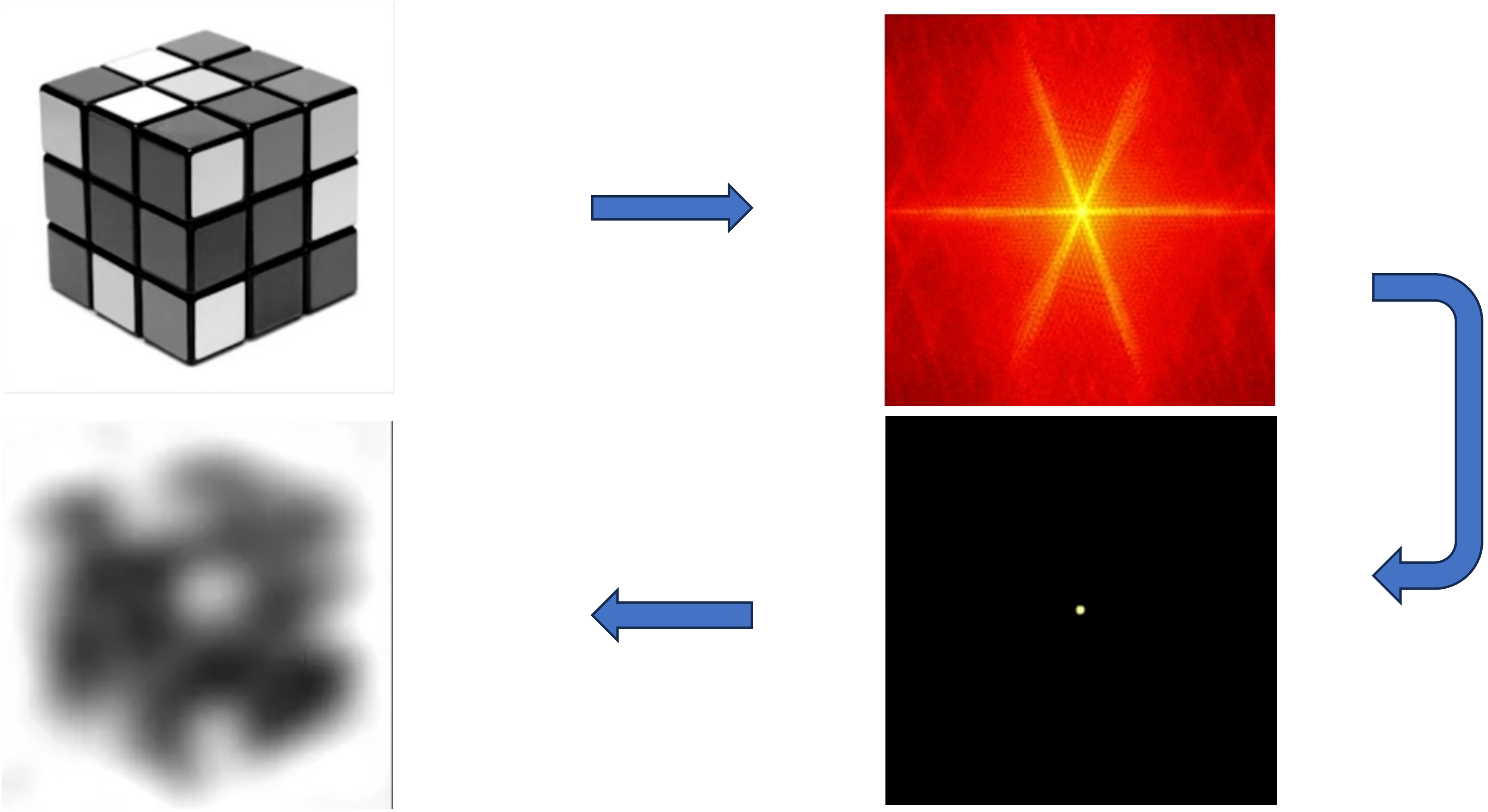


Image Filtering in the Frequency Domain

Application to image filtering: The low pass filter

f(m, n) *log(|F(p, q)|)*



Min Max

Color to show the magnitude

Image Filtering in the Frequency Domain

Application to image filtering: The high pass filter

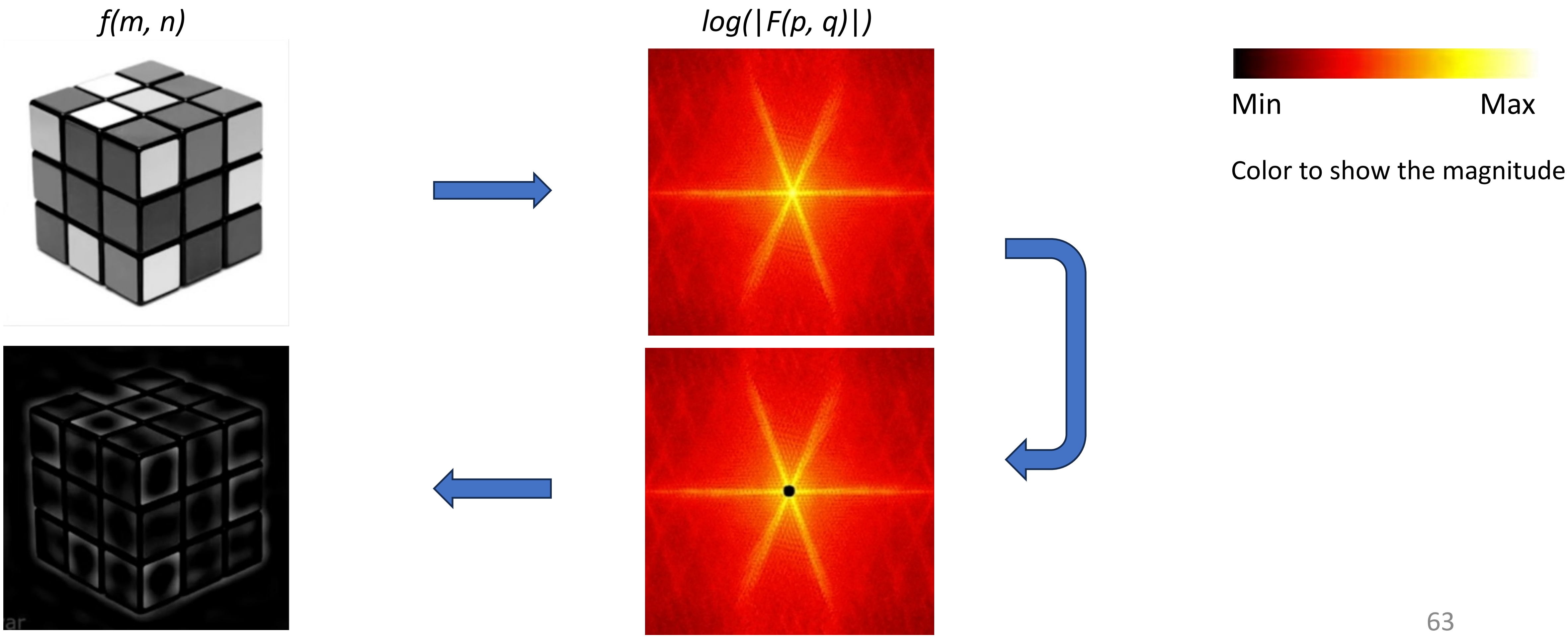
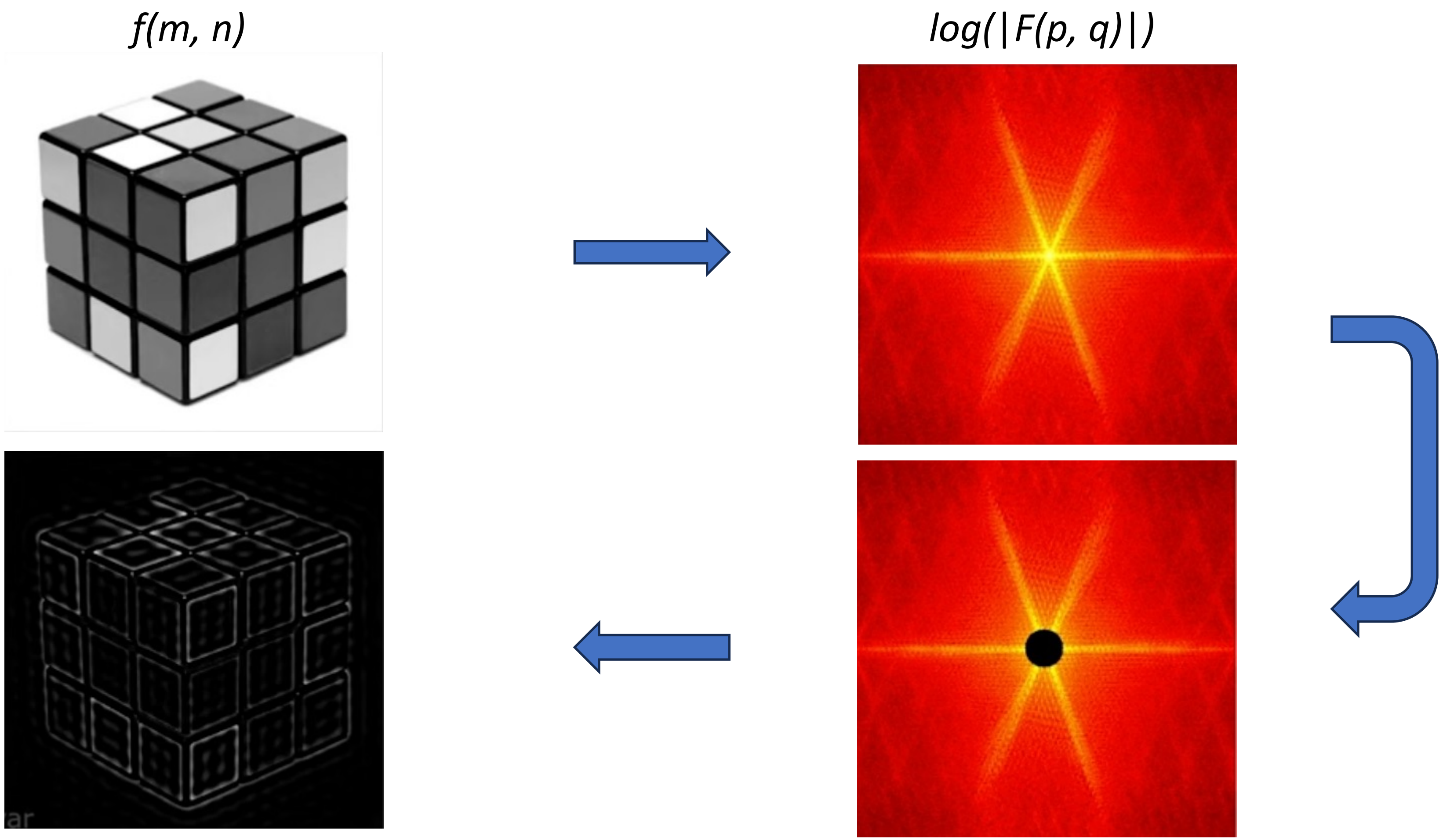


Image Filtering in the Frequency Domain

Application to image filtering: The high pass filter

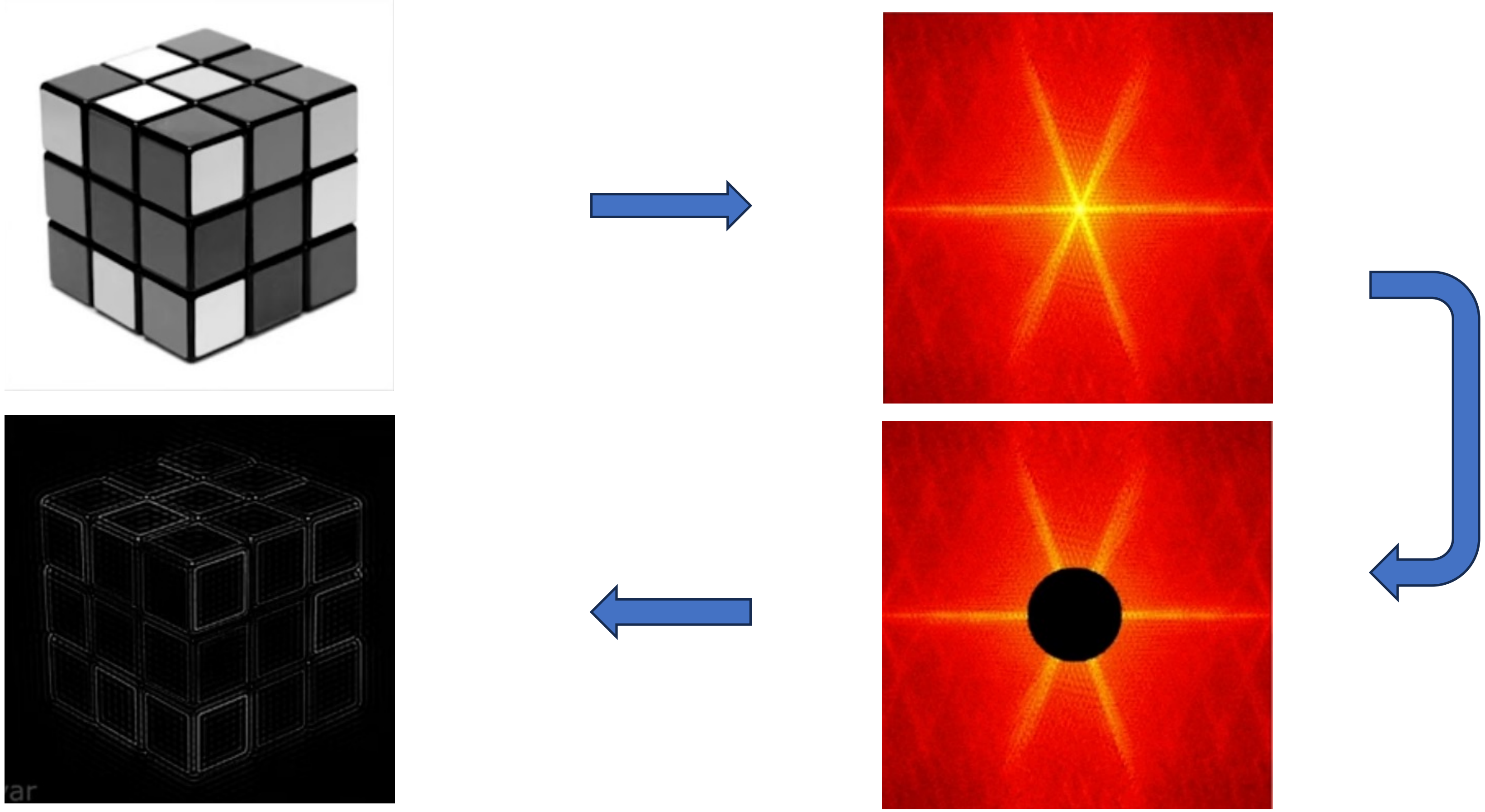


Color to show the magnitude

Image Filtering in the Frequency Domain

Application to image filtering: The high pass filter

f(m, n) | *log(|F(p, q)|)*



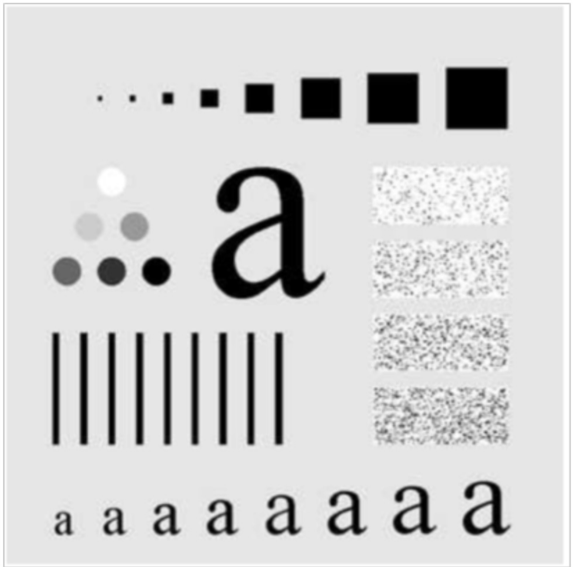
Color to show the magnitude

Image Filtering in the Frequency Domain

Low-pass filter seen so far is called Ideal Low-pass filter (ILPF):

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

The point of transition between $H(u, v) = 1$ and $H(u, v) = 0$ is called the **cut-off frequency**.



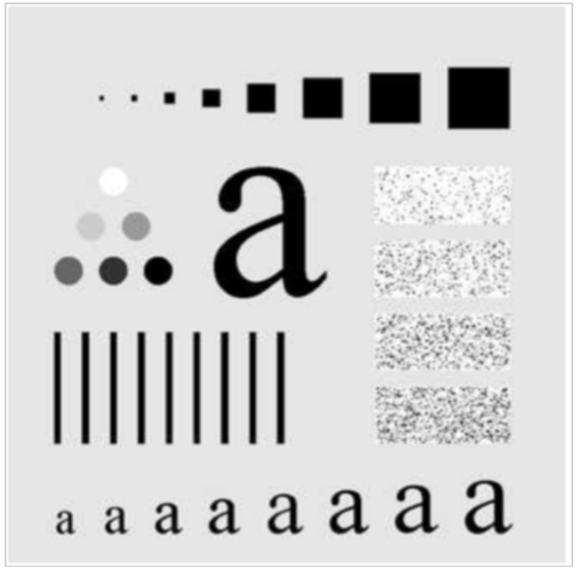
ILPF with cut-off frequency = 60

Image Filtering in the Frequency Domain

Low-pass filter seen so far is called Ideal Low-pass filter (ILPF):

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

The point of transition between $H(u, v) = 1$ and $H(u, v) = 0$ is called the **cut-off frequency**.



ILPF with cut-off frequency = 30

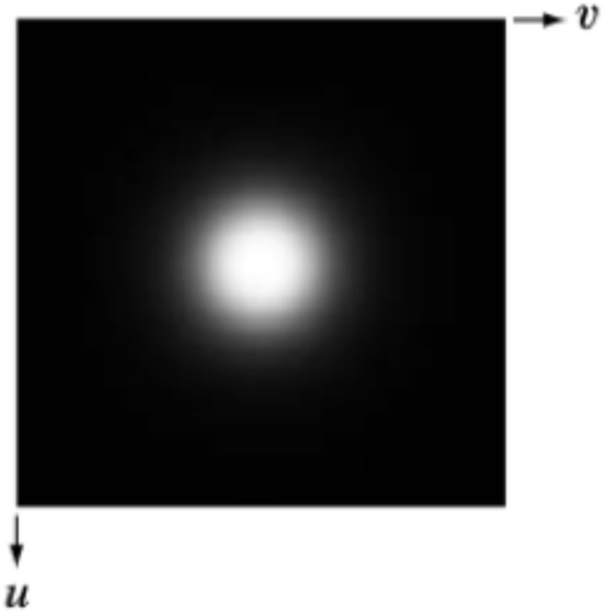
Image Filtering in the Frequency Domain

The butterworth Low-pass filters

$$H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}}$$

n is the order of the filter.

D_0 is the cutoff frequency.



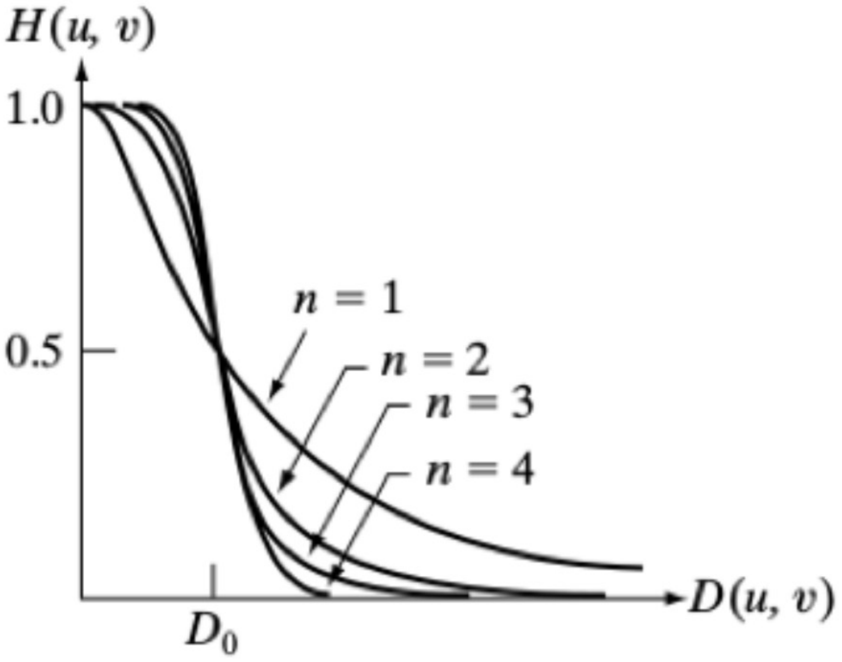
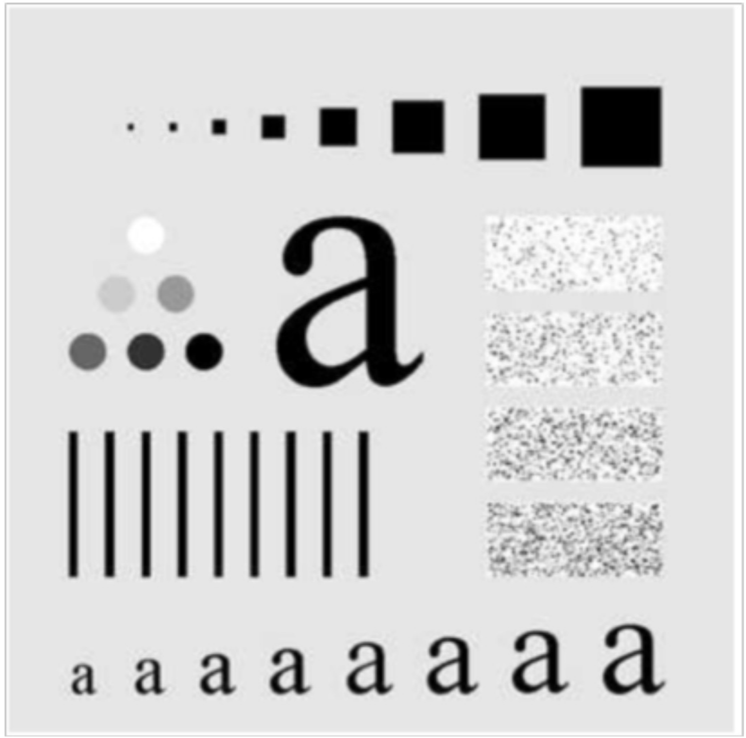


Image Filtering in the Frequency Domain

The butterworth Low-pass filters

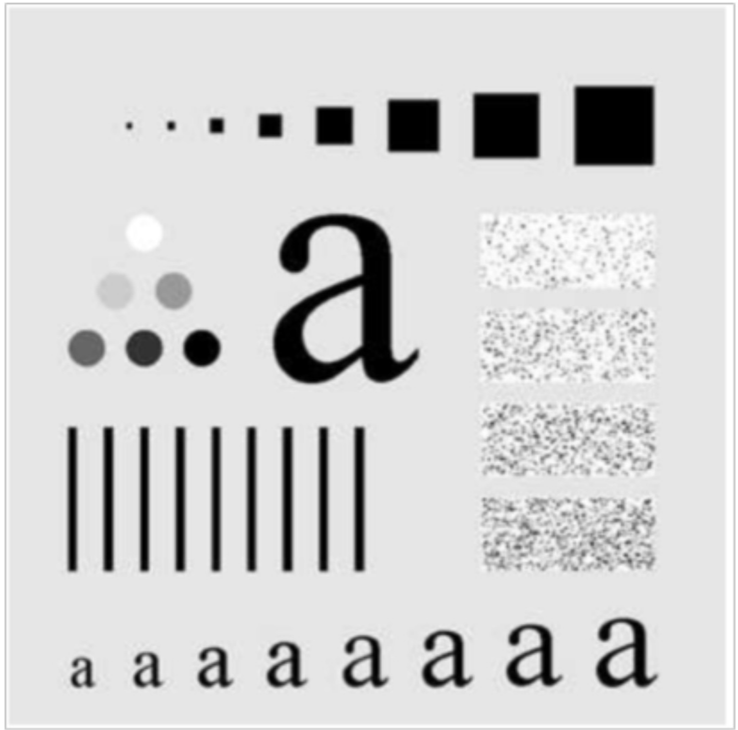


Order of the filter (n) = 2.

Cutoff frequency (D_0) = 60.

Image Filtering in the Frequency Domain

The butterworth Low-pass filters



Order of the filter (n) = 2.

Cutoff frequency (D_0) = 30.

Image Filtering in the Frequency Domain

Application to image filtering: The Gaussian filtering

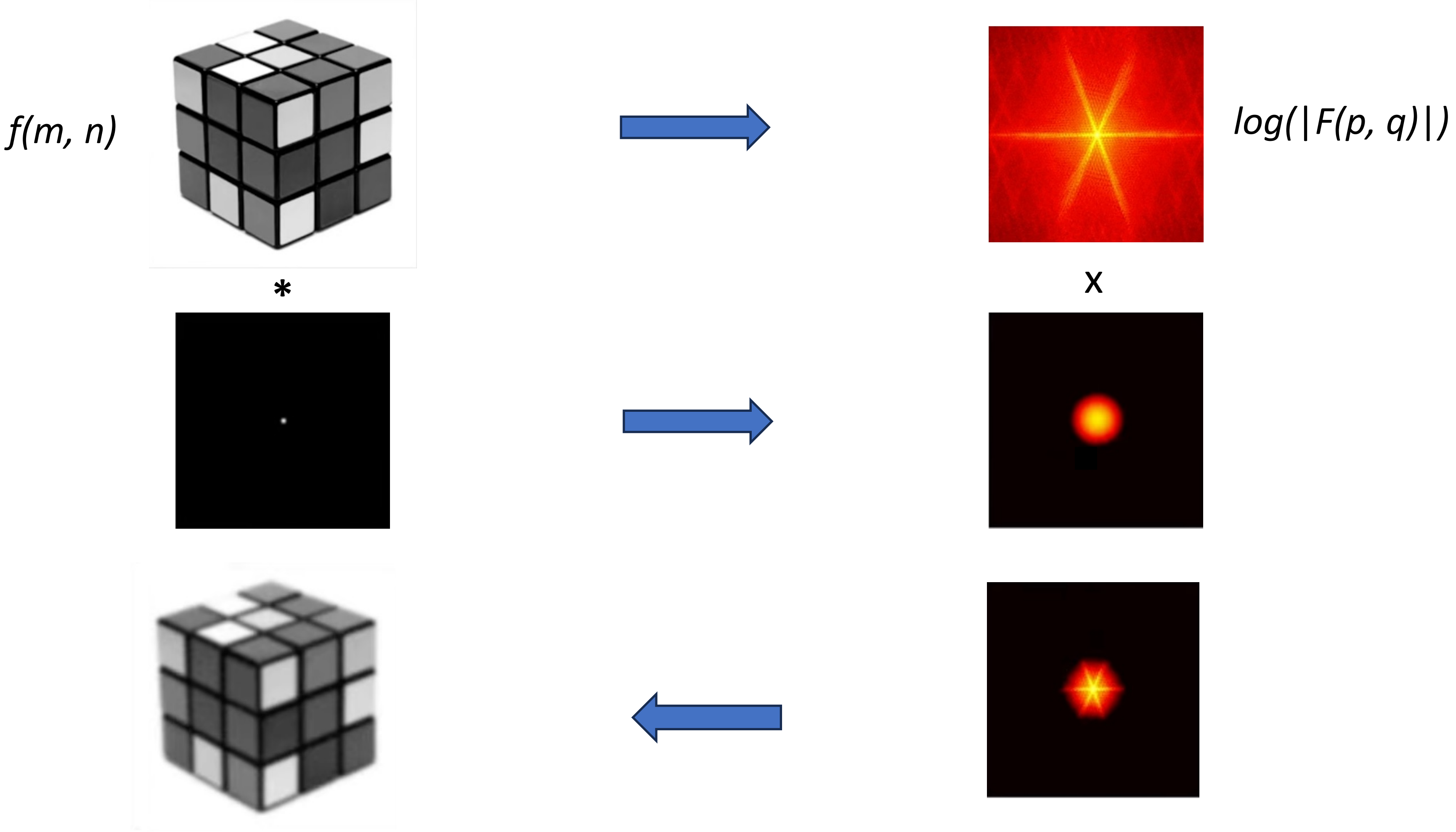


Image Filtering in the Frequency Domain

Application to image filtering: The Gaussian filtering

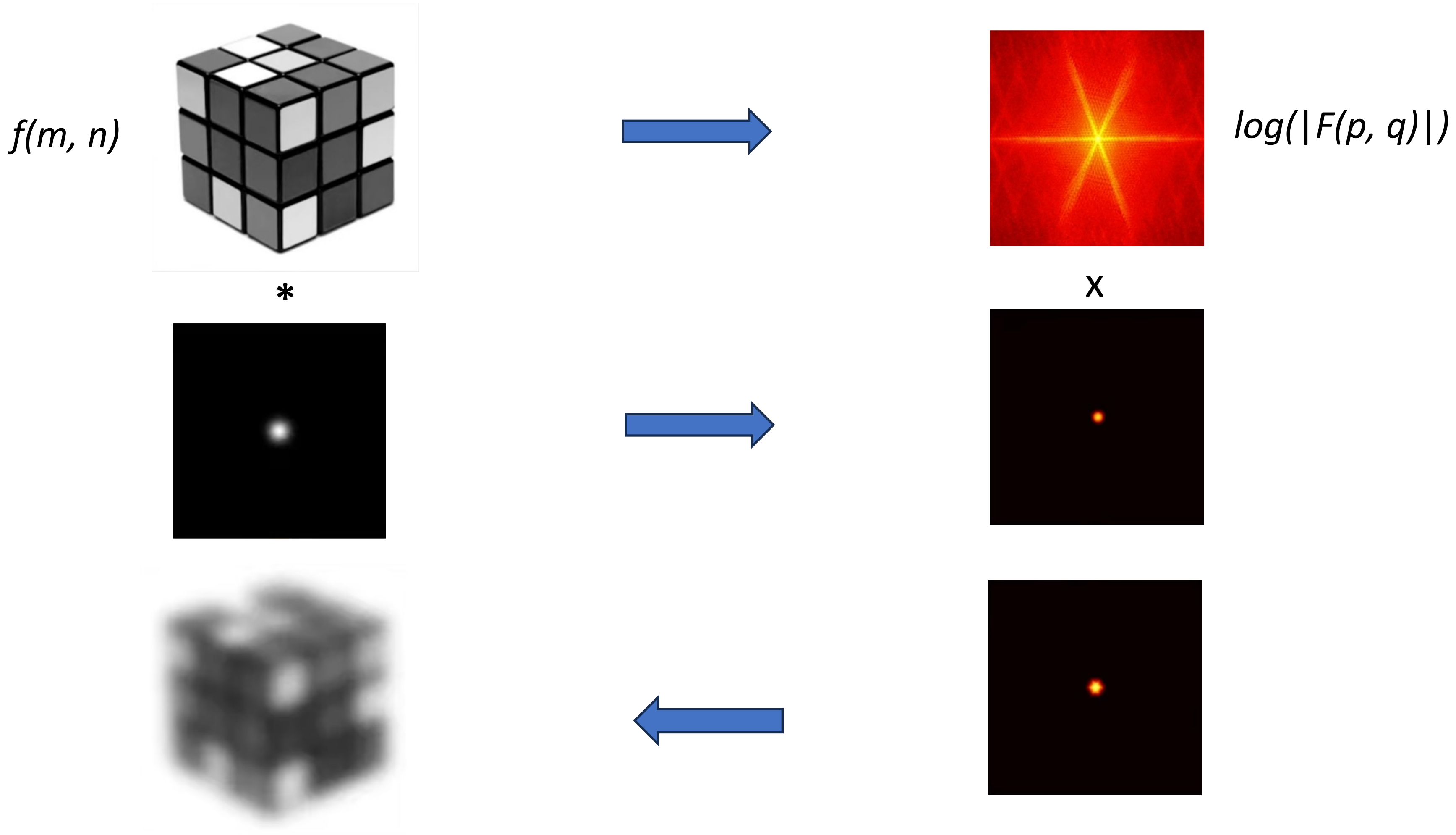


Image Filtering in the Frequency Domain

Application to image filtering: The Gaussian filtering

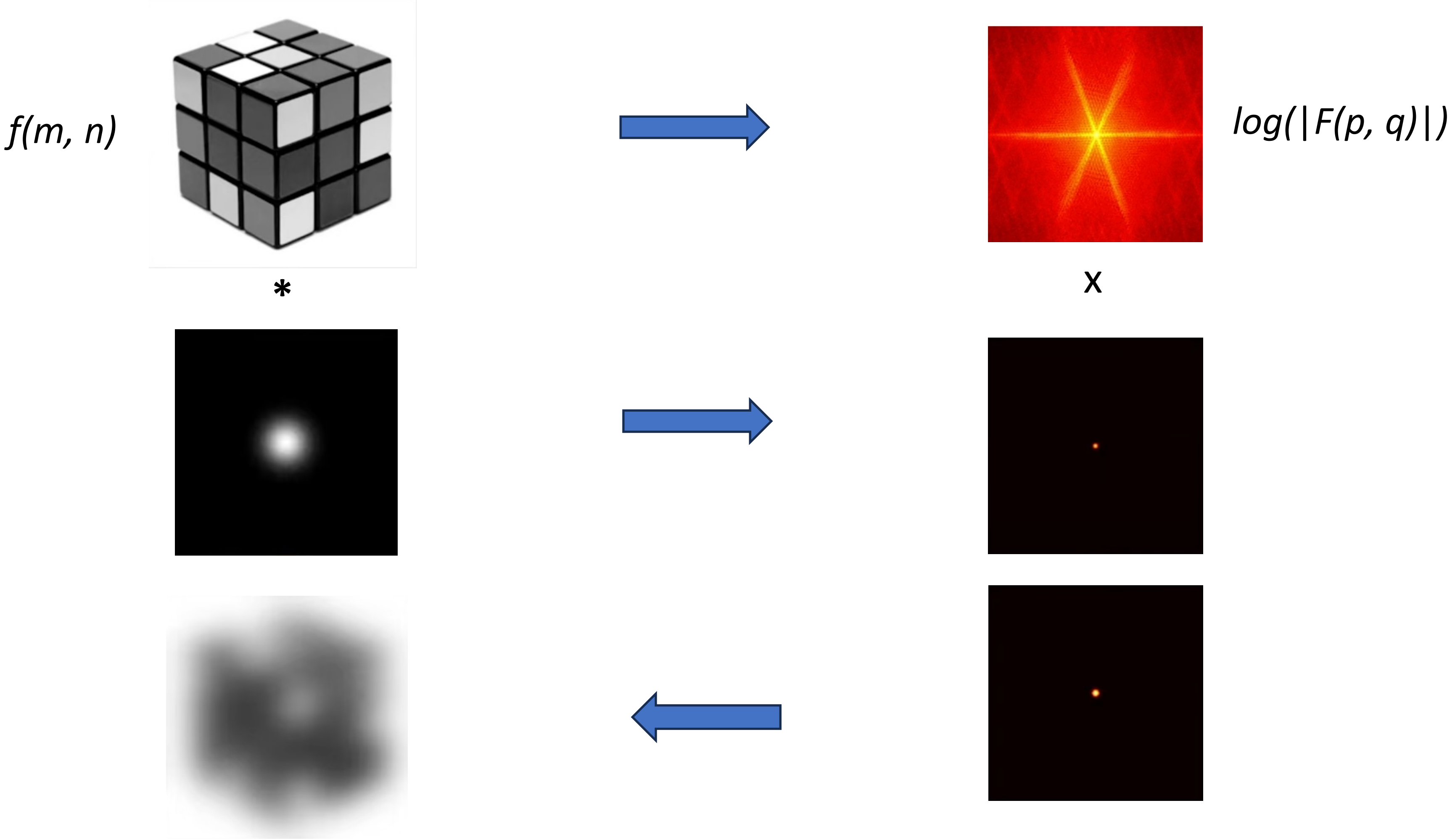
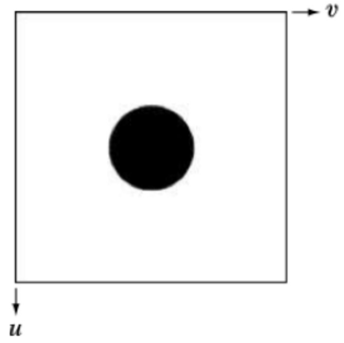
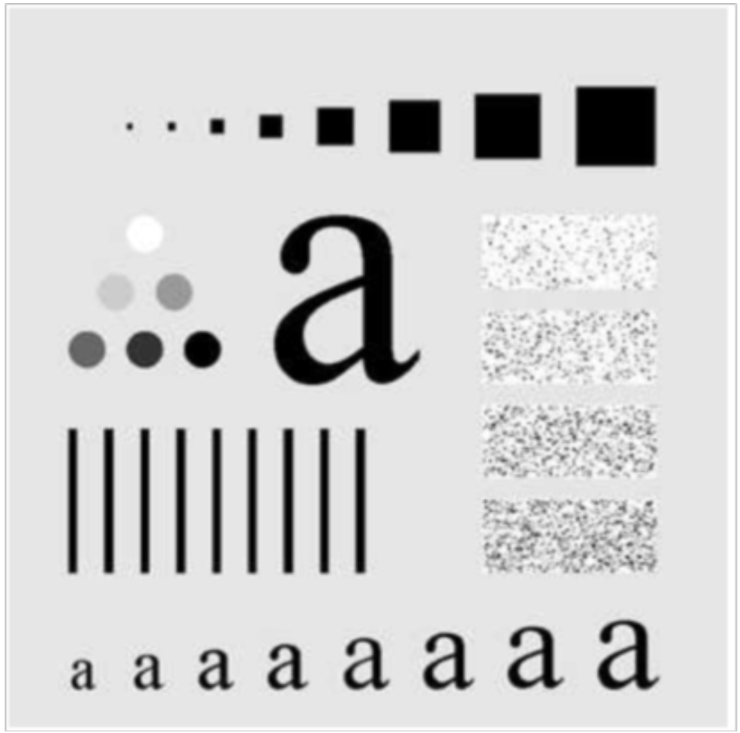


Image Filtering in the Frequency Domain

High-pass filter seen so far is called Ideal High-pass filter (IHPF):



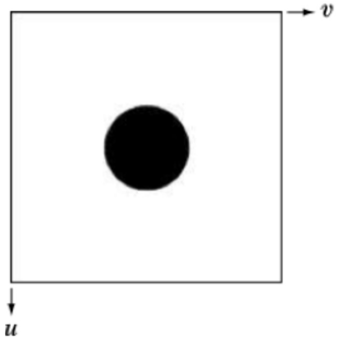
$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$



IHPF with cut-off frequency = 60

Image Filtering in the Frequency Domain

High-pass filter seen so far is called Ideal High-pass filter (IHPF):



$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \leq D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

IHPF with cut-off frequency = 30

Image Filtering in the Frequency Domain

The butterworth High-pass filters

$$H(u,v) = \frac{1}{1 + [D_0/D(u,v)]^{2n}}$$

n is the order of the filter.

D_0 is the cutoff frequency.

BHPF with cut-off frequency = 60

Image Filtering in the Frequency Domain

The butterworth High-pass filters

$$H(u, v) = \frac{1}{1 + [D_0/D(u, v)]^{2n}}$$

n is the order of the filter.

D_0 is the cutoff frequency.

BHPF with cut-off frequency = 30

Image Filtering in the Frequency Domain

In all of the previous examples, we have used only the magnitude.

Filtering has been performed considering only the magnitude.

What about the phase?

In many cases, the phase is more important than the amplitude to preserve the visual information.

Image Filtering in the Frequency Domain

In general,

- The amplitudes of the sinusoids determine the intensities in the image.

- The phase is a measure of displacement of the various sinusoids with respect to their origin.

    Carry much of the information about where discernable objects are located

Image Filtering in the Frequency Domain

Original image

Apply DFT

Set all the phases to 0

Preserve the magnitude

Reconstruct the image using the IDFT

Reconstructed image

Réf. : *Signal reconstruction from Fourier transform sign information.* S. Curtis, S. Lim, A. Oppenheim. 1984.
https://dspace.mit.edu/handle/1721.1/4243

Image Filtering in the Frequency Domain

Original image

Apply DFT

Preserve the phases

Using an average of magnitudes computed on a set of images

Reconstruct the image using the IDFT

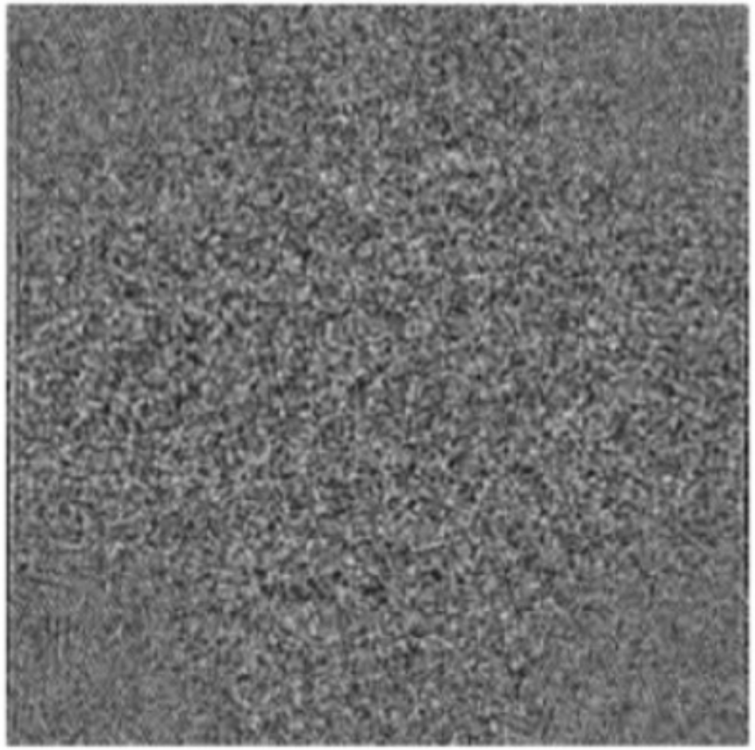
Reconstructed image

Réf. : *Signal reconstruction from Fourier transform sign information.* S. Curtis, S. Lim, A. Oppenheim. 1984.
https://dspace.mit.edu/handle/1721.1/4243

Image Filtering in the Frequency Domain

Example of image reconstruction using only the phase

The original image

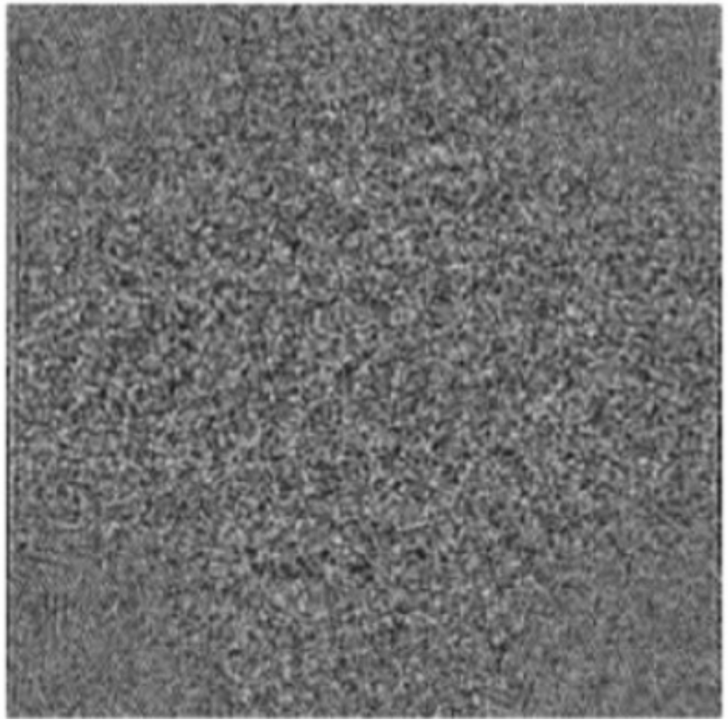


The phase

Image Filtering in the Frequency Domain

Example of image reconstruction using only the amplitude

The original image



The phase

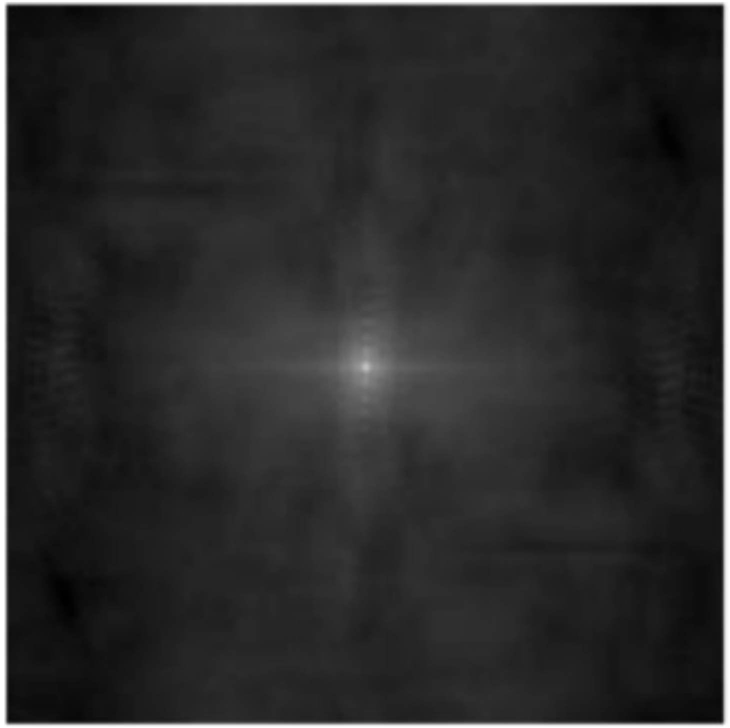


Image Filtering in the Frequency Domain

Example of image reconstruction using the phase and another amplitude

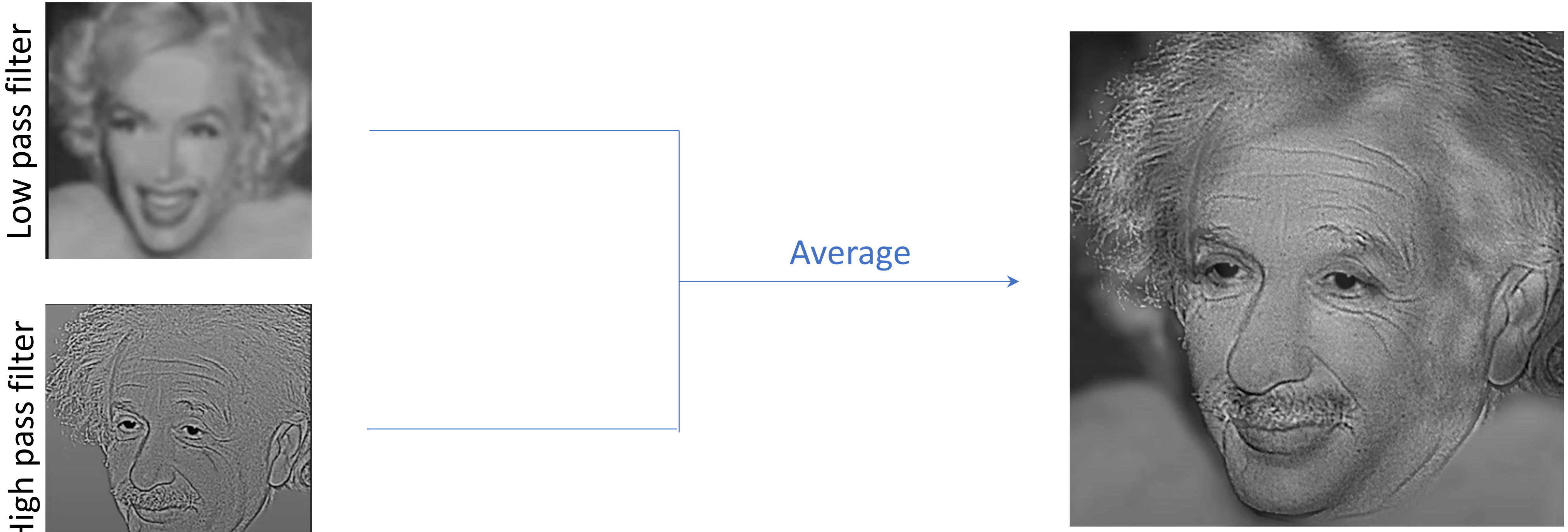
Image from which we compute the amplitude

Image from which we compute the phase

Reconstructed Image

Image Filtering in the Frequency Domain

Hybrid images

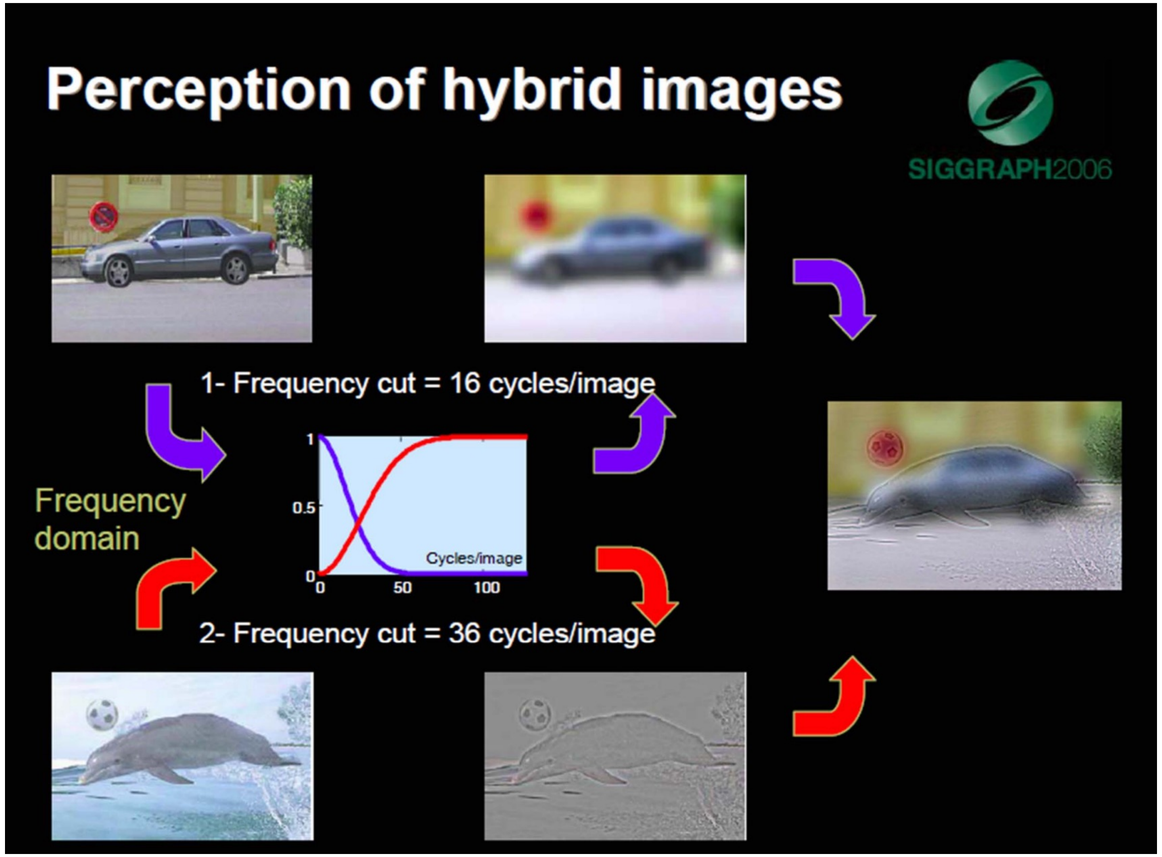


Hybrid image

Réf. : *Hybrid images.* A. Oliva, A. Torralba, P. Schyns. ACM Transactions on Graphics, 2006.
https://stanford.edu/class/ee367/reading/OlivaTorralb_Hybrid_Siggraph06.pdf

Image Filtering in the Frequency Domain

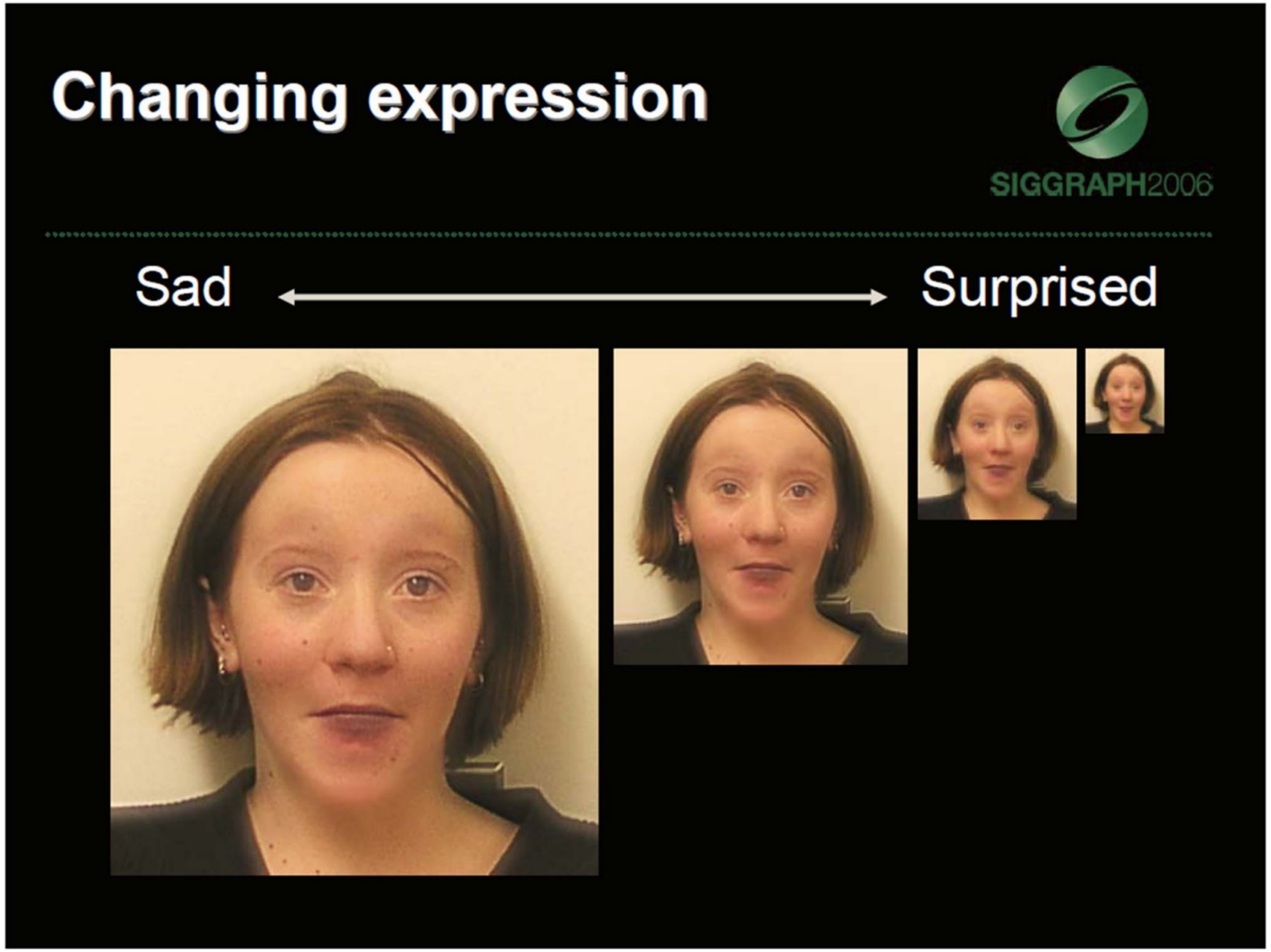
Hybrid images



Réf. : *Hybrid images.* A. Oliva, A. Torralba, P. Schyns. ACM Transactions on Graphics, 2006.
https://stanford.edu/class/ee367/reading/OlivaTorralb_Hybrid_Siggraph06.pdf

Image Filtering in the Frequency Domain

Hybrid images



Aude Oliva & Antonio Torralba & Philippe G Schyns, SIGGRAPH 2006